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"Evaluating Brain Alignment as a Training Signal for Language Models"

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Evaluating Brain Alignment as a Training Signal for Language Models

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Abstract

Language-model representations can predict human brain responses to language recorded with functional magnetic resonance imaging (fMRI). That result comes from frozen models, so it leaves the training question open. Does adding brain responses as an auxiliary objective alongside knowledge distillation (KD) give a smaller student model an advantage that comes specifically from brain-response content? The responses are either recorded from people or synthesized from text by a model trained on recorded responses. The question splits into five claims. Measurement is predicting recorded responses from a frozen model; manipulation is changing the student with a brain-response objective; attribution traces that change to brain-response content rather than to any dense target; transfer is improved prediction of independently recorded responses; and utility is improvement on a practical endpoint. Each claim needs its own comparator, and none establishes another. Measurement holds. On an isolated-sentence dataset and a spoken-story dataset from one deeply sampled participant, frozen representations predicted held-out responses beyond simple stimulus properties, and in the story data beyond an architecture-matched untrained model. Training on recorded responses then failed at attribution. Against a control trained on the same responses permuted across sentences, correct pairing gave no reliable advantage across nine participants, the inference unit this claim requires. Mean scores on a downstream language endpoint favored the recorded responses, while the predictivity change they were meant to pay off proved unstable across seeds. Training on synthetic responses reached manipulation and went no further. The student learned the target and stayed changed after the training-only components were removed, at language quality inside a tolerance set in advance. Transfer beyond ordinary KD was not demonstrated at the participant level, and the text-derived control was too poorly matched to attribute any difference to brain-response content. The measurement assays and the manipulation checks passed, so a broken pipeline does not explain these negative findings. No end-to-end positive control was available, however, so the sensitivity of the full intervention-to-transfer path remains unestablished. Neither route establishes brain alignment as a usable training signal; both leave open a benefit from other targets, interventions, or populations. The contribution is the separation that keeps a valid positive result at one stage from being read as support for the training claim.

1 Introduction

Brain responses to language could help train a smaller language model (LM). In knowledge distillation (KD), a smaller student model learns to reproduce the output distribution of a larger teacher [1], but comparable output behavior does not require the two models to organize information identically in their middle layers. When several model representations support text prediction about equally well, brain responses could favor one of them over the others. Privileged information, a target available only during training, can shape which representation the student settles on without changing the inputs available at deployment [2, 3]. Here that target is either a recorded functional magnetic resonance imaging (fMRI) response or a synthetic brain response generated from text by a separate model trained to predict fMRI responses.

Whether brain responses and model representations share information at all is first a measurement question. We assess brain alignment by fitting an encoding model that predicts recorded fMRI responses from frozen LM representations, then evaluating its predictions on stimuli withheld from fitting [4, 5]. The resulting score is a property of the whole assay, and only partly of the LM. It depends on which participants were scanned, which stimuli they read or heard, how their responses were aggregated, and how reliably each was measured. The score also depends on the data split, the nuisance features (position, length, and static lexical features), the class of readout that maps representations to responses, and the unit over which uncertainty is computed. Apparent model–brain correspondences can be produced or reordered by data splits that scatter locally related stimuli across fitting and evaluation, by simple stimulus features, and by analysis choices [6, 7]. In this manuscript, *controlled brain predictivity* refers to brain alignment that survives the nuisance, split, and control-model checks specified in Section 3.2.

A positive brain-alignment score is computed from frozen representations, with no training intervention. It therefore does not establish that training with brain responses can change the representations a student retains. Even if such a change occurs, the score does not establish three further claims, namely (1) that the change is caused specifically by brain-response content rather than generic geometric properties of the target or the extra optimization pressure of any auxiliary loss; (2) that it transfers to independently recorded fMRI responses; and (3) that it improves a practical endpoint such as a downstream task, at comparable LM quality.

Prior work does not resolve whether brain responses can help train a smaller model. Studies that turn recorded brain responses into a training target report changes in model representations or task behavior in several language and speech settings [8–11]. Whether such a change comes from brain-response content is a separate question, because privileged-information and feature-distillation research shows that a dense training-only target can help a student for reasons unrelated to its source [2, 3].

We therefore ask directly whether training with brain responses gives a fixed smaller student an incremental advantage over a matched control, at comparable LM quality and with inference taken over participants. We evaluate each component of that question against the evidence that component alone requires. The model representations must show controlled brain predictivity. How far the smaller model's controlled brain predictivity falls below its teacher's, beyond what worse LM quality explains, bounds the opportunity that training with brain responses could recover. Training with recorded fMRI responses must be compared with training on the same responses permuted, which preserves the response values and breaks their pairing with stimuli. For synthetic brain responses generated from text, we ask whether the declared 50-component linear projection of the training target adds measurable information about recorded fMRI responses, and whether the student learns that target. We also ask whether training produces changes that remain after removing the prediction head, an auxiliary output layer added only for training. Training with synthetic brain responses must be compared with training that uses a similarly constructed text-derived auxiliary target, at comparable LM quality and at the participant level. Because both arms share the same student-training regime and differ only in how the auxiliary target was constructed, this comparison tests whether any difference is specific to brain-response content. We must also determine whether training improves prediction of independently recorded fMRI responses beyond ordinary KD, and performance on an external task. If one of these requirements is not met, that limits what the claims depending on it can establish, and leaves the other claims untouched.

In the systems tested, this study established controlled brain predictivity and a measurable change in the student from synthetic-response training, but not a brain-specific training advantage. The distilled student had lower controlled brain predictivity than its teacher, but worse LM quality remained a competing explanation for this gap. Training with recorded fMRI responses did not produce a reliable participant-level benefit in the tested regimes. On the practical endpoint, the recorded-response arm performed better on average than its permuted-response control. We did not report uncertainty for that paired comparison, and the controlled-predictivity prerequisite was unstable across random seeds, so brain-specific utility was not demonstrated. For synthetic brain responses generated from text, the 50-component projection of the training target did not clear the threshold fixed in advance for that linear assay. No other claim tested here depends on that assay. The target was nevertheless learnable on the training corpus, and training changed the student's retained representations while preserving LM quality within the prespecified tolerance. The direct participant-level comparison between training with synthetic brain responses and ordinary KD found a near-zero difference in prediction of recorded fMRI responses. The available text-derived auxiliary target was not sufficiently comparable to attribute any difference between the synthetic and text-derived arms specifically to brain-response content.

This study contributes controlled measurement of brain predictivity alongside LM quality, and participant-level tests of training with recorded fMRI responses. It also separates five questions about training with synthetic brain responses generated from text, asking whether a declared linear projection of the target was measurable, whether the student learned it, whether the change survived removal of the prediction head, whether an adequate comparator existed, and whether the change transferred to independently recorded fMRI responses. The conclusion is limited to the models, English stimuli, datasets, and participants tested, and to the targets, objectives, controls, readouts, comparators, and endpoints used. The study does not provide a successful brain-response-guided distillation method, but neither does it show that brain responses can never help train a smaller model.

2 Background and Related Work

Four adjacent literatures bear on the question: brain-encoding studies [4, 5], brain-response-guided optimization [8–11], privileged-information and feature-distillation work [2, 3], and model-compression studies [12]. This thesis keeps five claims apart. *Measurement* is predicting recorded responses from a frozen model, *manipulation* is changing the student with a brain-response objective, *attribution* is tracing that change specifically to brain-response content, *transfer* is improvement on independently recorded responses, and *utility* is improvement on a practical endpoint. None of these claims establishes another. Predicting recorded brain responses does not establish that training with those responses changes the student, and a change in the student leaves attribution, transfer, and utility open. A brain-specific training advantage is an incremental benefit of brain-response content beyond a matched control, at comparable LM quality and measured at the correct biological inference unit.

2.1 What controlled brain predictivity establishes

Controlled brain predictivity is the measurement claim. It tests whether representations extracted from a frozen LM contain information useful for predicting recorded fMRI responses to held-out stimuli. An encoding model is fitted to predict responses from those representations on one set of stimuli and evaluated on a separate set. This held-out evaluation tests whether prediction generalizes beyond the fitting stimuli [4, 5].

Three controls are required before predictive performance can be attributed to learned parameters rather than to simpler alternatives or to leakage between fitting and evaluation sets. First, a nuisance-only model predicts recorded fMRI responses from stimulus properties such as position, length, and static lexical features. Comparing it with a full model that also includes the tested representations measures what those representations add. Second, an architecture-matched untrained LM retains the tokenizer, dimensionality, and architecture of the trained model but lacks its learned parameters. This comparison tests whether predictive performance depends on learned parameters rather than the architecture alone. Third, contiguous or blocked splits keep locally related samples from being scattered across fitting and evaluation sets, reducing inflated scores caused by dependence or leakage between them [6]. Together, these controls narrow what a positive score licenses to held-out predictive value added by learned LM representations under the declared assay. That assay is the stated combination of stimuli, response definition, readout, and controls.

Even with these controls, controlled brain predictivity remains bounded by the declared assay. A linear readout tests whether recorded responses can be predicted from a linear combination of LM representation dimensions; it does not test every form of information that might be present. Participant-specific responses measure predictivity for an individual, whereas averaged responses measure predictivity of a group average. The inference unit determines the scope of uncertainty, because many voxels or repeated folds from one participant do not provide evidence about many independent participants. Measurement reliability and the number of averaged repetitions set a ceiling on what any model can predict [13]. The stimulus distribution and the selected brain regions bound it further. Controlled brain predictivity therefore supports a bounded claim about held-out prediction under the declared assay, not a shared biological mechanism or causal equivalence [7]. Sections 3.2 and 3.5 specify how this assay was instantiated.

2.2 Why brain-response targets might help a student

Ordinary KD trains a student to reproduce the teacher's output distribution, but this objective does not uniquely determine how the student organizes information in its middle layers. Different bases, subspaces, and feature organizations can support similar token predictions. This leaves room for an auxiliary training target to favor one representation over another without necessarily improving or degrading the language objective.

Brain-response targets can fill that role as privileged information, data available during training but unnecessary at deployment [2]. The student keeps the LM objective used in ordinary KD, while an auxiliary objective favors representations from which the training target can be predicted. Recorded fMRI responses vary with the participant and session beyond what the text fixes. Synthetic brain responses generated from text instead carry only the structure learned by a fixed generator, a model trained beforehand to map text to brain responses. Because each synthetic response is determined by the text and generator, it cannot provide example-specific information that varies independently of them. It may nevertheless present generator-learned structure in a form that is easier for the student to learn or recover. That possibility is a hypothesis motivated by privileged-information and feature-distillation results rather than a consequence of the target's deterministic construction [2, 3].

Evidence that the auxiliary objective affected training does not establish a brain-specific training advantage. Reduced target loss can show only that components present during training learned to predict the auxiliary target, and those components can be discarded afterwards. The retained student, the model kept once those components are removed, may therefore be unchanged. Manipulation is checked in two parts, both against the ordinary KD arm. Target uptake requires a new prediction model, fitted after training, to recover the target better from the retained student's frozen representations than from that baseline. Retained-student movement separately requires parameter or representation differences from the same baseline that remain after components used only during training are removed. Neither uptake nor movement establishes attribution, transfer, or utility. Sections 3.3 and 3.4 specify the objectives, gradient paths, and baselines for these checks.

2.3 What prior brain-response-guided training leaves unresolved

Prior work assigns brain responses two distinct roles, post-training evaluation and training supervision. Post-training encoding studies use recorded fMRI responses to evaluate model representations after the model has been changed for another reason. Narrative-summarization tuning can increase brain-predictivity scores without using brain responses during training, while model scaling can increase naturalistic brain alignment under the studied assay even when matched-size instruction tuning does not [14, 15]. These findings show that brain predictivity can change because of the training objective or model scale alone; they do not estimate the value of brain responses as training targets. Brain-response-guided studies instead use recorded brain responses or model-generated brain-response predictions during training [8–11]. Those studies report that training with these responses can change model representations, brain-predictivity scores, or selected task outcomes. However, the claim supported by each result depends on its comparator, evaluation data, and inference unit. Table 1 compares selected designs and what each leaves unsettled.

Positive studies span language, speech, and vision; they do not estimate a single shared effect of brain-response training. Speech models adapted with recorded fMRI responses have improved brain-response prediction and selected semantic or auditory tasks, with some gains extending to participants

Table 1: Prior model interventions that train with brain responses, and their reported outcomes.

Study	Intervention and relevant result	Why it does not settle this study's question
Bilgin et al. [16]	A cosine loss against participant-specific fMRI responses, together with a token loss, updates upper-layer adapters; held-out encoding, new-participant readouts, and a model-specific external task improve.	No permuted or matched text-derived target, smaller student, or ordinary KD arm.
Merlin, Moussa, and Toneva [10]	Participant-specific fMRI-response, stimulus-language, or joint objectives update adapters; brain-response training outperforms stimulus tuning on more linguistic-probe comparisons.	Target geometry and difficulty are not matched between the brain-response and stimulus-language objectives; no participant-held-out transfer or compression comparison.
Merlin and Toneva [17]	Complementary objectives add or remove brain-predictive information; a permuted-response control and matched LM loss accompany changes in linguistic probes.	No smaller student; the readout used to add or remove information may itself carry linguistic variance not specific to brain-response content.
Pirlot et al. [18]	A monkey-V1-response regularizer trains a vision model; permuted targets reproduce the clean-accuracy gain, while real targets improve moderate-attack robustness.	Only the robustness endpoint distinguishes brain-response content in that design.

whose responses were not used in adaptation [8, 19, 20]. Studies of text-model interventions report gains on multilingual, probe, commonsense, and reasoning endpoints [9, 10, 16, 21]. Training with temporally precise electrocorticography responses has improved prediction within the same modality, while training with slower fMRI responses has transferred to electrocorticography prediction for new participants and stimuli after fitting a new readout [11, 22]. In one of these studies the brain-prediction gains were scored on stimuli drawn from the same distribution as the adaptation data, so they do not establish transfer across stimulus distributions [20]. These studies differ in model, training target, endpoint, personalization, and transfer setting, so their positive results do not establish the incremental value of brain responses for a fixed smaller student.

Privileged-information and feature-distillation studies provide a competing explanation for any benefit from training with brain responses. A dense training-only target may produce an improvement by emphasizing distinctions useful to the student or, as a so-far untested hypothesis, by making optimization easier, regardless of whether the target contains brain-response content. The vision study in Table 1 is one such case [18]. The value of a dense training-only target depends on the information it carries, the student's capacity, and how the teacher and student representations are matched [2, 3]. Ruling that explanation out therefore requires an arm whose target matches the brain-response target in density and learnability but is derived from text alone, under the same LM objective.

Evaluating the student after training must also meet requirements from measurement-validity research. Post-training brain predictivity must be measured with leakage-resistant splits and nuisance controls [6, 7]. It must also be measured at an inference unit appropriate to the claim, because the number that enters a test is the number of independent units, not the number of observations pooled from them [23]. Compression studies show that pruning or quantization method and model scale can change brain predictivity without brain-response training [12]. An intervention comparison must therefore separate the effect of the brain-response objective from changes in LM quality and must evaluate the retained student with a fresh measurement rather than rely on the value of its own training loss.

The unresolved question posed before any new experiment is therefore narrower than whether training with brain responses can change a model. For a fixed smaller student trained by ordinary KD, do recorded fMRI responses or synthetic brain responses generated from text provide an incremental, brain-specific advantage over the appropriate control, which is permuted responses for recorded targets and a matched text-derived target for synthetic targets at comparable LM quality?

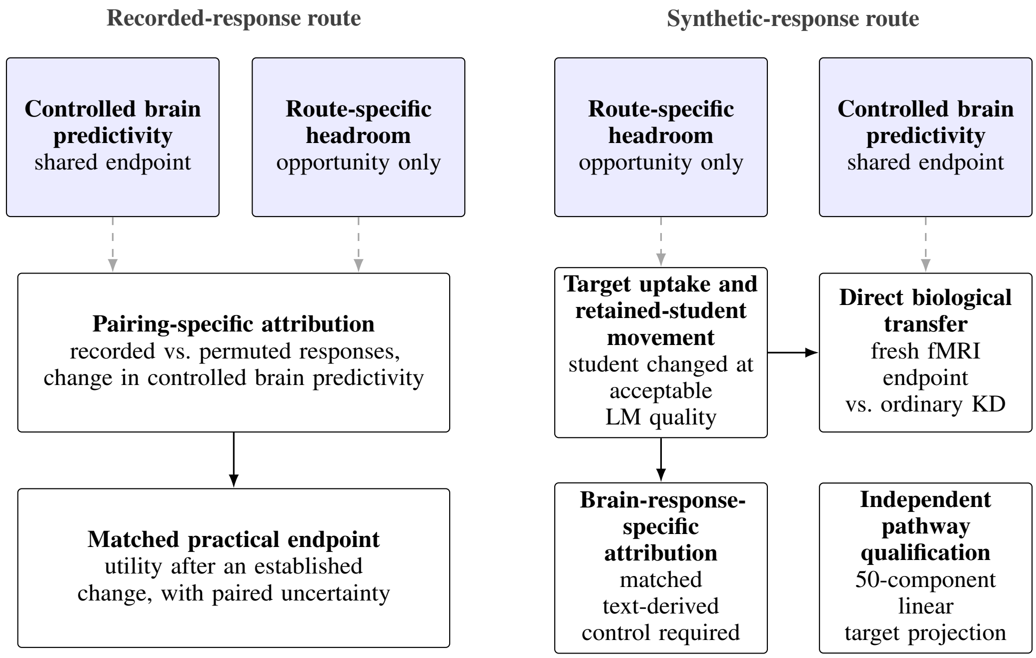


Figure 1: Claim dependencies for recorded- and synthetic-response training. Dashed arrows mark a claim that supplies context for the claim it points to; solid arrows mark a claim that must pass first.

2.4 Evidence criteria for a brain-specific training advantage

Figure 1 organizes the evidence by what each claim depends on, rather than as one fixed sequence every claim must pass through. Route-specific headroom describes only the opportunity to improve the shared endpoint, not the validity of a later paired intervention contrast. The two routes then carry different claims, namely pairing-specific attribution and utility in the recorded-response route, and manipulation, transfer, and attribution in the synthetic-response route, alongside an independent pathway qualification.

No single comparator supports every claim in Figure 1. An architecture-matched untrained LM tests whether controlled brain predictivity depends on learned parameters. An ordinary KD arm isolates the incremental effect of adding a brain-response objective to the same student-training regime, while comparable LM quality limits generic quality differences. The route-appropriate attribution control depends on the training target. In the recorded-response route, the attribution control compares correctly paired with permuted responses, testing whether the stimulus–response pairing matters while preserving the response values and their marginal distribution. In the synthetic-response route, the attribution control compares the synthetic brain-response target with a text-derived auxiliary target matched on properties that we fix or calibrate in advance, before interpreting the target-arm outcomes. These properties include geometry, scale, baseline predictability, remaining headroom, and the early gradient magnitude it exerts. Target uptake and retained-student movement, defined in Section 2.2, can explain how two target interventions differed, but they are outcomes of those interventions rather than conditions the comparator must satisfy.

A freshly fitted readout on independently recorded fMRI responses tests biological transfer; successful recovery of the optimized training target cannot substitute for this endpoint. A practical-utility claim requires improvement over its route-appropriate control on a prespecified endpoint; when the claim is that improved controlled brain predictivity pays off practically, it additionally requires that the predictivity improvement itself be reliably established first. The direct transfer comparison, the synthetic-response arm against ordinary KD, is interpretable without the text-derived auxiliary control. The auxiliary control is additionally required only when the claim attributes a difference to brain-response content.

Each result therefore licenses only the claim that depends on it. A failed manipulation check limits the interpretation of a null intervention result, but it does not invalidate a positive independently measured endpoint contrast. The 50-component linear target-projection assay tests whether the declared projection of the training target adds controlled predictivity of recorded fMRI responses beyond nuisance and the frozen row-twin control, which is the same target rows with their sentence pairing broken. The assay qualifies that linear pathway alone. It does not gate any manipulation, transfer, or attribution claim, so Figure 1 draws it as an isolated box with no arriving or departing dependency arrows. An unresolved comparison must remain unresolved rather than be converted into a null result, and failure under one readout, cohort, or target must not be generalized beyond that scope. Section 3 maps each claim to its data, training arms, estimand, comparator, endpoint, and inference unit; Section 4 reports the corresponding verdicts.

3 Methods

Section 2.4 defines the evidence criteria for converting controlled brain predictivity into a brain-specific training claim. This section instantiates those criteria. For each claim it names the data, the intervention or assay, the comparator, the estimand, the endpoint, and the inference unit. An estimand is the specific quantity or effect that an analysis is designed to estimate. Table 2 maps each claim role to its design tuple. A role may appear in more than one row when different datasets, response constructions, or population units define different estimands.

Appendix A maps these scientific roles to the experiments that settled them and records the interpretations that were superseded during review.

3.1 Study design, datasets, and data partitioning

Two datasets provide the recorded fMRI responses. The Tuckute condition-B dataset [24] contains 1,000 isolated English sentences, responses from 10 participants, and five fixed left-hemisphere language regions of interest (ROIs). The dataset supports the controlled regional predictivity assay and the distillation-headroom analysis. The same sentences later support the recorded-response route and the Tuckute endpoints of the target-projection and saved-student biological-transfer assays. The LeBel corpus [25] provides continuous story stimuli for the controlled naturalistic voxelwise predictivity assay. That assay uses participant UTS03, reliability-screened voxels (Appendix B), and 20 stories divided into five held-out-story folds. Holding out complete stories keeps temporally dependent samples from the same story on one side of the evaluation boundary. The resulting estimand concerns held-out-story prediction within UTS03 and does not support population inference across people.

All recorded responses come from previously published, publicly released datasets and are analyzed here as secondary data; no new human data were acquired for this thesis. Participant recruitment, informed consent, and ethical approval were the responsibility of the originating studies and are reported there [24, 25]. The released data carry only study-assigned participant identifiers, which are the identifiers used throughout.

The Tuckute responses can be constructed at two response-aggregation levels. Participant averaging combines the five Tuckute training participants declared in Appendix B into one sentence-by-ROI matrix. These five are the participants the original Tuckute encoding fit used, so the averaged target follows the published construction rather than a subset selected here. This matrix describes the shared stimulus-locked response, but it cannot support participant-level inference. Participant-specific construction instead keeps one response matrix per recorded participant. One participant is excluded because that participant's required five-ROI response matrix is incomplete, leaving nine valid participants. Contrasts can therefore be aggregated within participant before inference across the nine tested participants. Response coordinates also differ in spatial resolution. An ROI coordinate summarizes responses within a predefined language region, whereas a voxel coordinate is a finer spatial measurement. Table 2 pairs these constructions with their comparisons and inference units.

Each training and evaluation role draws on a declared text and response source. The recorded-response intervention trains on Tuckute sentences and their recorded responses. The synthetic-response intervention instead trains on 95,999 WikiText-103 sentences [26]. A separate 1,999-sentence WikiText split supports language-quality evaluation and synthetic-target recovery. For the target-projection assay, the fixed synthetic brain-response target is generated solely from the Tuckute sentence text by TRIBE [27], a published text-to-brain response model (Appendix B). Recorded fMRI responses enter only the assay's held-out evaluation. Saved students, the students kept from each trained arm

Table 2: Claim roles and their primary designs.

Stage	Scientific role	Primary comparison and endpoint	Inference unit
Measurement	Controlled regional predictivity assay	Trained representations versus nuisance and architecture-matched untrained LM; held-out ordinary and unique R^2	Fixed sentence set; five contiguous folds
Measurement	Controlled naturalistic voxelwise predictivity assay	Trained versus untrained representations beyond nuisance; held-out-story voxelwise unique R^2	Story blocks within one deeply sampled participant
Opportunity	Distillation-headroom analysis	Teacher versus post-KD student declared-layer brain predictivity, interpreted with held-out LM quality	Model condition; training seed for stochastic arms
Linear pathway	Target-projection assay	Synthetic target versus nuisance and frozen row twin; held-out recorded-response unique R^2	Participant on a fixed sentence set
Manipulation	Synthetic-target recovery assay	Synthetic-response versus ordinary KD arms; held-out WikiText target-recovery R^2; saved block-permuted variants are sensitivity checks	Training seed
Manipulation	Retained-student movement audit	Synthetic-response versus ordinary KD arms; retained parameter or representation change at comparable LM quality	Training seed; fixed target sample for geometry
Attribution	Recorded-response intervention, participant-averaged estimand	Correctly paired versus permuted recorded responses; fresh averaged-response unique R^2	Shared stimulus fold; seeds are technical repetitions
Attribution	Recorded-response intervention, participant-specific estimand	Correctly paired versus permuted recorded responses; fresh participant-response unique R^2	Participant after fold and seed aggregation
Attribution	Comparator-adequacy audit and attribution assessment	Synthetic-response versus text-derived auxiliary-control arms; training parity, target parity, and baseline learnability, then the synthetic-response minus text-derived arm contrast	Seed for trained models; fixed target sample for matching checks
Manipulation	Saved-student target-retention analysis	Synthetic-response versus ordinary KD students; held-out synthetic-target recovery	Training seed
Transfer	Composed-path diagnostic	Aligned versus frozen row-twin targets and synthetic-response versus ordinary KD students; held-out recorded-response unique R^2 through the composed path	Participant
Transfer	Saved-student biological-transfer assay	Synthetic-response versus ordinary KD arms; fresh participant-response unique R^2	Participant
Utility	Practical-utility evaluation	Recorded-response arm versus permuted-response control at approximately matched language quality; held-out out-of-domain to in-domain perplexity ratio	Technical seed; fixed corpora and procedure

after the training-only components are discarded, remain frozen; the saved-student biological-transfer assay fits a fresh readout to recorded Tuckute responses for each evaluation. Pereira sentences and LeBel story transcripts provide out-of-domain language-quality endpoints for the practical-utility evaluation [25, 28]. Recorded LeBel fMRI responses are confined to the naturalistic assay.

The data partitions preserve these separate roles. The principal Tuckute roles use five contiguous outer partitions. Each partition holds out 200 sentences and leaves the other 800 as its training complement. For the controlled regional predictivity assay, distillation-headroom analysis, target-projection assay, and saved-student biological-transfer assay, this boundary separates fresh-readout

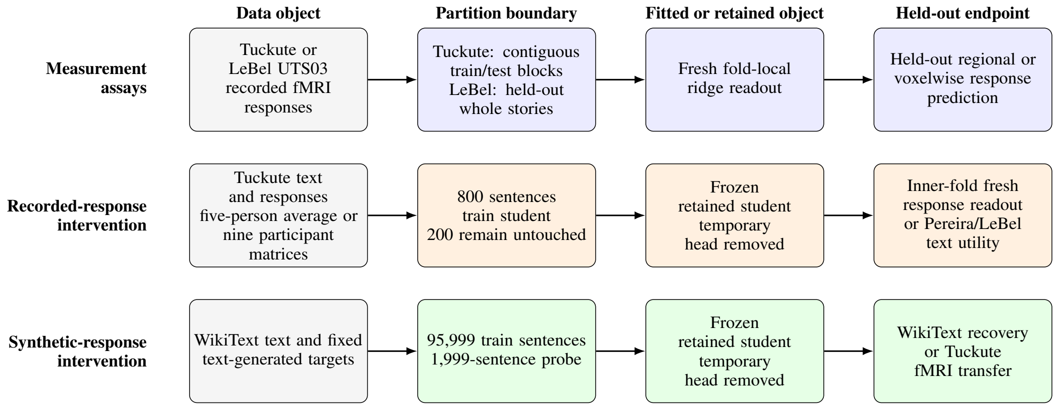


Figure 2: Data and partition paths used by the designs in Section 3.

fitting from evaluation. The recorded-response intervention uses the outer boundary differently. It optimizes the student on the 800-sentence training portion, which excludes the held-out 200 sentences. Within that held-out block, five inner folds separate fresh-readout fitting from readout evaluation. The synthetic-response route uses its two WikiText partitions for student training and target-recovery evaluation. Within that route, recorded Tuckute responses enter only the later target-projection and saved-student biological-transfer assays. Fold-dependent transformations are fitted on the corresponding training portion. The target-projection assay has one declared exception. It reuses the target mean and scale that were fixed once during target construction, before any fold split existed. Its subsequent target principal component analysis (PCA) and all readout transformations remain fold-local. Appendices B and D report the exact inclusion, preprocessing, partition, and nesting definitions.

Figure 2 separates the measurement assays from the recorded-response and synthetic-response interventions. The measurement assays fit readouts without changing an LM; the two interventions first produce a retained student and then evaluate it on held-out data. The arrows denote data use, training, or evaluation rather than causal influence.

3.2 Measurement assays and opportunity diagnostics

Both controlled brain predictivity assays address the same question. Does a contextual LM representation improve held-out prediction of recorded fMRI responses beyond specified nuisance features? Each assay first fits a nuisance-only readout containing its declared low-level and static features. The full readout adds a contextual representation block. The contextual and static blocks are each reduced to the same PCA rank, so their widths match. Let y_{ij} denote observation i of response coordinate j, N the nuisance block, X the contextual block, and $\mathcal{T}$ one held-out fold. For any readout design M, ordinary held-out R^2 is

$$R_j^2(M) = 1 - \frac{\sum_{i \in \mathcal{T}} (y_{ij} - \widehat{y}_{ij}^M)^2}{\sum_{i \in \mathcal{T}} (y_{ij} - \bar{y}_{\mathcal{T}j})^2}. \tag{1}$$

A value of one denotes perfect prediction, zero matches a prediction that always uses the held-out mean, and a negative value means that the fitted readout predicts worse than that baseline. The nuisance-only and full scores are $R_j^2(N)$ and $R_j^2([N, X])$. Their cross-validated semipartial, or unique, score is

$$\Delta R_j^2(X) = R_j^2([N, X]) - R_j^2(N). \tag{2}$$

A positive value means that the contextual LM representation improves held-out response prediction beyond the declared nuisance features.

Figure 3 separates this nuisance comparison from the two additional controls. One control uses an untrained LM with the same architecture and rank. In the naturalistic assay, this control uses the matched nuisance construction and tests whether learned parameters matter. The regional assay as executed here recomputed the static nuisance block for each model, so its trained-minus-untrained

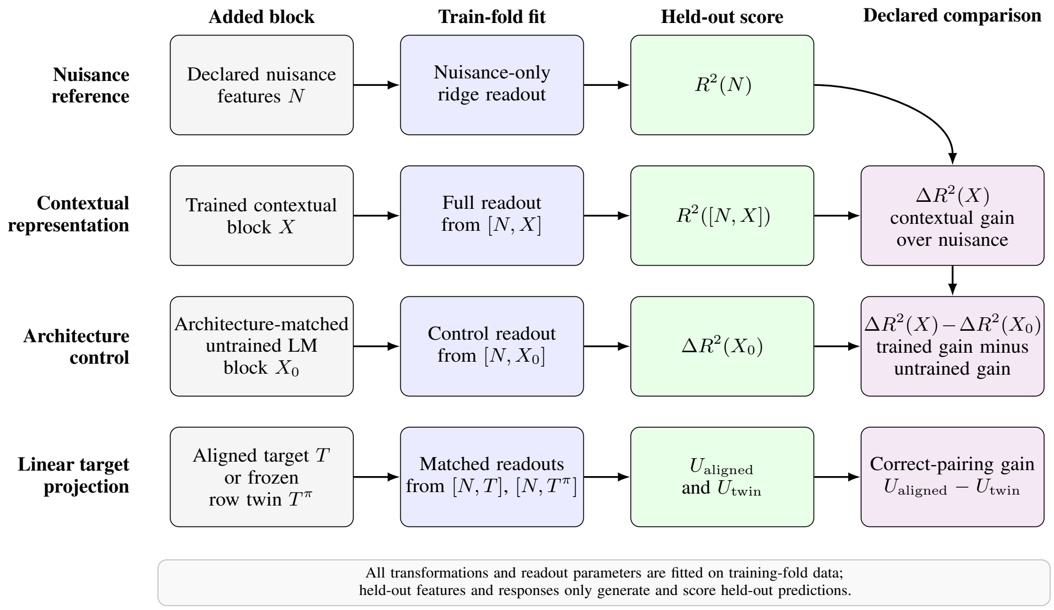


Figure 3: Fold-local readout comparisons used by the measurement assays.

gap does not isolate learned parameters against a fixed nuisance block. The frozen row twin, constructed with the target-projection assay below, instead tests whether correct sentence–target pairing matters. Repeating the contextual comparison across the teacher and students supplies the scores used by the distillation-headroom analysis. Released Tuckute reliability estimates provide descriptive context for the regional scores.

The controlled assays define the biological measurement context, but they do not show how much predictivity a particular teacher–student route leaves available to recover. The distillation-headroom analysis addresses that separate opportunity question by applying the same Tuckute assay at the declared layers. Its reference models are a pretrained GPT-2 Medium (`gpt2-medium`) teacher, an untrained GPT-2 Small architecture-matched to the students, and an off-the-shelf pretrained GPT-2 Small (`gpt2`). The analysis also evaluates two GPT-2 Small students after ordinary KD, one from random initialization and one from pretrained weights. Let $\overline{\Delta R^2}(m, s)$ denote the declared-ROI and fold average of Equation 2 for model m and technical seed s. For teacher m_T, the seed-specific headroom is

$$\mathcal{H}(m, s) = \overline{\Delta R^2}(m_T) - \overline{\Delta R^2}(m, s). \tag{3}$$

Equation 3 measures the teacher predictivity that the tested student has not recovered, so zero headroom means their averaged scores are equal under this assay. For Figure 7a, let A_0, A_S, and A_T denote the corresponding untrained-model floor, student, and teacher scores. The floor-anchored retention is

$$\rho' = \frac{A_S - A_0}{A_T - A_0}. \tag{4}$$

Equation 4 reports the fraction of the teacher-to-untrained-floor interval retained by the student. Values of zero and one place the student at the floor and teacher anchors, respectively. Each brain-predictivity score is reported alongside held-out LM quality because quality may explain part of the difference. A cross-family analysis instead uses tokenizer-comparable bits per byte and relates $\overline{\Delta R^2}(m)$ directly to model quality; it does not define teacher-relative headroom across unrelated model families. The analysis measures route-specific headroom under the tested conditions. It does not identify model size, architecture, initialization, training history, distillation, or language quality as the cause of that headroom.

The 50-component linear target-projection measurability assay, called the target-projection assay from here on, examines the fixed synthetic brain-response target that TRIBE generates from text. It asks whether the declared fold-local PCA representation of that target predicts recorded Tuckute responses on the same 1,000 sentences. For each participant, the assay fits three otherwise matched response readouts. The first contains nuisance features only. The second adds the aligned synthetic target. The third adds its frozen row-twin control. This control applies one deterministic permutation to complete target rows within each outer block and has no fixed points. It preserves the exact target-row multiset within that block but breaks sentence–target pairing. Let T denote the aligned target block and T^{π} its frozen row twin. Both use the same train-fold target-PCA basis and the same nuisance and readout procedure. Write $R^2_{u,f,j}(M)$ for Equation 1 applied to participant u, fold f, and response coordinate j. For the J declared response coordinates, define

$$
\begin{aligned}
U_{\text{aligned}}(u,f) &= \frac{1}{J}\sum_{j=1}^{J}\left[R^2_{u,f,j}([N,T]) - R^2_{u,f,j}(N)\right], \\
U_{\text{twin}}(u,f) &= \frac{1}{J}\sum_{j=1}^{J}\left[R^2_{u,f,j}([N,T^{\pi}]) - R^2_{u,f,j}(N)\right].
\end{aligned}
\tag{5}
$$

These two quantities apply the same nuisance subtraction to the correctly paired target and its row-twin control. Their difference can therefore test whether correct sentence–target pairing adds predictive value beyond the information retained by the target rows after permutation.

Let $A(u)$ denote participant-level measurability beyond nuisance. Let $Q(u)$ denote the additional predictive value of correct sentence–target pairing. The two participant-level estimands, averaged over the $K = 5$ folds, are

$$
\begin{aligned}
A(u) &= \frac{1}{K}\sum_{f=1}^{K} U_{\text{aligned}}(u,f), \\
Q(u) &= \frac{1}{K}\sum_{f=1}^{K}\left[U_{\text{aligned}}(u,f) - U_{\text{twin}}(u,f)\right].
\end{aligned}
\tag{6}
$$

Equation 6 first averages the aligned target's unique predictivity across folds and then averages the aligned-minus-twin pairing contrast across those same folds. Computing both quantities separately for each participant makes the participant, rather than a fold or response coordinate, the biological inference unit. The assay therefore tests participant-level predictive measurability on this fixed stimulus set. It does not establish causal mediation, brain-response-specific attribution, or stimulus-population generality.

All assays in this subsection use the same general held-out readout protocol. A fresh ridge readout is fitted to frozen features for every evaluated model or target. Negative R^2 values are retained when predictions are worse than the held-out-mean baseline. All learned preprocessing, response centering, regularization selection, and readout fitting follow the training-partition rule in Section 3.1. Appendix B specifies the exact inputs, preprocessing, and declared standardization exception. Appendix D specifies the formulas, aggregation, and inference procedures.

3.3 Recorded-response intervention

Within each outer fold and technical seed, the recorded-response arm and permuted-response control start from the same student state and use the same sentences. They also share the ordinary KD objective, optimizer schedule, loss weight, and training budget. Only the pairing between recorded-response rows and their sentences differs. Both arms distil a Qwen2.5-1.5B teacher into a Qwen2.5-0.5B student using low-rank adaptation (LoRA) [29]. The teacher remains frozen. Using the partitions defined in Section 3.1, each run trains on the 800 training sentences of its partition. Five outer folds and three technical seeds repeat this paired design.

Ordinary KD provides the shared language-preservation objective. Let $\mathcal{L}_{\text{KD}}$ denote the temperature-scaled, teacher-to-student Kullback–Leibler (KL) divergence over all nonpadding tokens. Let $\mathcal{L}_{\text{R}}^{(a)}$ denote the mean-squared auxiliary response loss for arm a. The complete training objective is

$$
\mathcal{L}_{\text{train}}^{(a)} = \mathcal{L}_{\text{KD}} + \lambda_{\text{R}}\mathcal{L}_{\text{R}}^{(a)}, \qquad a \in \{\text{recorded}, \text{permuted}\}. \tag{7}
$$

Equation 7 states that both paired arms are anchored by the same teacher-defined language objective while receiving an additional response-based training signal. The arm identity changes the response pairing, not the language objective or the strength assigned to the response loss. An ordinary KD arm without the response loss is only a secondary comparison here, because its language quality need not match that of the paired arms. Appendix C specifies the exact temperature, masking reduction, optimizer, schedule, and parameter states.

The recorded-response arm uses the response vector paired with each sentence. The control applies the same loss to response rows permuted in contiguous sentence blocks. This permutation preserves the response values, their marginal distribution, and their within-block structure while disrupting the sentence–response correspondence. A random block permutation can leave some blocks in their original positions, so some sentences keep their own response rows. The participant-specific regime repeats this control with five permutation draws and averages their evaluated scores when forming the paired contrast. Appendix B gives the exact permutation construction.

Participant-averaged and participant-specific responses instantiate the same paired intervention but answer different questions. In the participant-averaged regime, each fold–seed run uses the fixed participant-averaged response construction defined in Section 3.1. Its contrast describes the effect of correct pairing for that shared stimulus-locked target. The participant-specific regime instead trains separate paired arms for each of the nine valid participants. Its contrast supports inference across the tested participants. Averaging responses before training is not equivalent to averaging participant effects after training. The participant-averaged contrast therefore cannot be interpreted as the cohort-mean intervention effect.

The auxiliary branch attaches to the masked-mean layer-12 student state through a temporary linear head W, and the auxiliary loss scores the prediction from W. In the primary LoRA regime, the pretrained base model and tied input/output embedding remain frozen, while W and the adapters are optimized. The auxiliary gradient can reach W and trainable adapters that produce the layer-12 state. It cannot reach later adapters through this branch, although those adapters can still receive gradients from $\mathcal{L}_{\mathrm{KD}}$. After training, the temporary head and teacher are discarded, the adapters are merged into the retained student, and that student is frozen for evaluation. Figure 4a shows where the ordinary KD and route-specific auxiliary objectives attach. Figure 4b shows that every post-training endpoint uses the retained student rather than the temporary head.

After each run, evaluation uses only the 200 held-out sentences of that partition. The endpoint of each run is the ROI-averaged unique R^2 defined by Equation 2. Let $U^{(a)}_{ufs}$ denote this score for participant u, outer fold f, technical seed s, and arm a. With $P = 5$ permutation draws indexed by p, the paired cell contrast and participant aggregate are

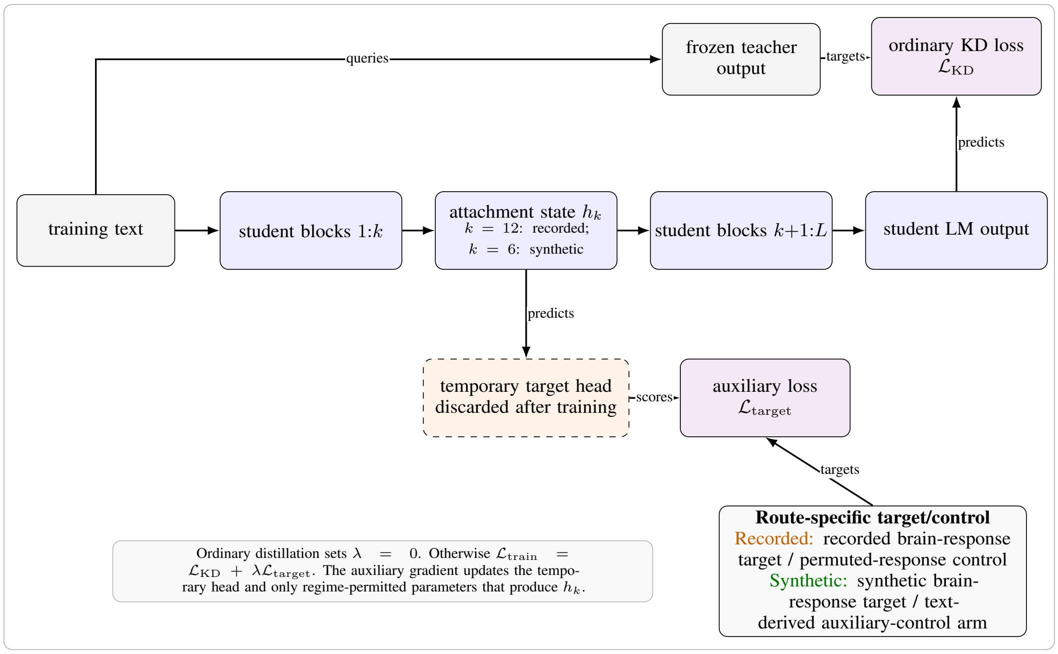
$$\begin{aligned} d_{ufs} &= U^{(\text{recorded})}_{ufs} - \frac{1}{P}\sum_{p=1}^{P} U^{(\text{permuted})}_{ufsp}, \\ e_u &= \text{median}_{f,s}\, d_{ufs}. \end{aligned} \tag{8}$$

Equation 8 first compares the recorded-response student with the average of its five matched permuted-response controls within each fold–seed cell. The median over fold–seed cells was fixed before the participant outcome run so that no single fold or seed defines a participant's effect. Averaging the nine participant effects estimates the intervention contrast in the tested cohort. The participant, not the underlying fold or seed, is the biological inference unit. A separate aggregation across the five shared outer folds checks whether the contrast is concentrated in particular stimuli. The participant-averaged analysis instead aggregates seeds within each shared fold. Appendix D specifies the intervals, tests, shared-fold checks, and sensitivity analyses.

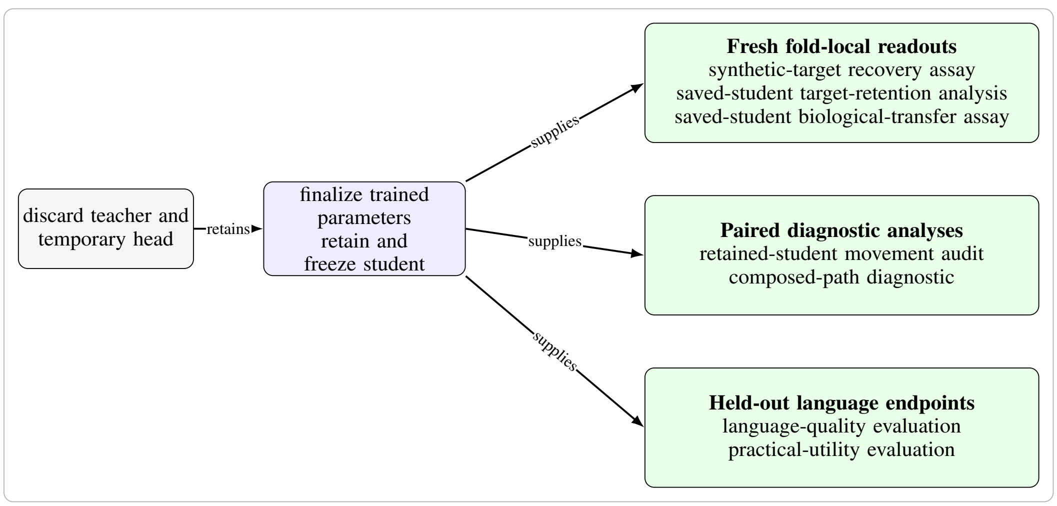
The frozen auxiliary head, increased LoRA capacity, a contrastive objective, naturalistic voxelwise evaluation, and full fine-tuning each retain the paired recorded-versus-permuted comparison while changing the target, target loss, language anchor, trainable subset, or evaluation scope. Appendix C defines these variants, and Section 4 reports their study-specific outcomes. None replaces the primary nine-participant contrast.

The practical-utility evaluation compares approximately quality-matched participant-averaged recorded-response and permuted-response students through the ratio of out-of-domain to in-domain perplexity on Pereira sentences and LeBel story text. For corpus d, arm a, and technical seed s, define

$$R_{d,a,s} = \frac{\mathrm{PPL}^{\mathrm{OOD}}_{d,a,s}}{\mathrm{PPL}^{\mathrm{ID}}_{a,s}}, \qquad \Delta R_{d,s} = R_{d,\text{recorded},s} - R_{d,\text{permuted},s}. \tag{9}$$



(a) Shared training core.



(b) Frozen-student evaluation.

Figure 4: Shared intervention training and frozen-student evaluation paths.

The ratio measures out-of-domain perplexity relative to the same student's in-domain perplexity. A negative $\Delta R_{d,s}$ favors the recorded-response arm. The prerequisite contrast in controlled brain predictivity is defined at the technical-seed level as

$$A_s = C_s^{\mathrm{recorded}} - C_s^{\mathrm{permuted}}, \tag{10}$$

where C_s is held-out controlled brain predictivity. This contrast tests whether the recorded-response arm produced the representational change whose downstream value the utility comparison is intended to assess. A brain-specific utility interpretation additionally requires two things: a reliable brain-response-specific change in the prerequisite contrast, and an uncertainty estimate for the paired endpoint contrast at the technical-seed unit. Section 3.5 states the formal aggregation and decision rules.

3.4 Synthetic-response intervention and downstream assays

The synthetic-response intervention asks whether fixed synthetic brain-response targets generated from text change a retained student beyond ordinary KD. Across six technical seeds, a frozen GPT-2 Medium teacher supervises a GPT-2 Small student in the synthetic-response, ordinary KD, and text-derived auxiliary-control arms. These arms share the initialized student, training text and order, tokenizer, ordinary KD objective, optimizer schedule, and training budget. The synthetic-response arm adds a fixed text-only TRIBE target, whereas the text-derived auxiliary-control arm mean-pools frozen teacher states and maps them through a fixed Gaussian projection. Both auxiliary targets are standardized to 20,484 coordinates and predicted from the student's layer-6 representation with a mean-squared-error loss. Training updates the student and a temporary prediction head; the head is then discarded and the retained student is frozen. Figure 4a locates the synthetic target and the text-derived auxiliary target within the shared training core. Figure 4b separates target recovery, retained-student movement, biological transfer, composed predictive overlap, and language evaluation after the temporary head has been removed. Appendices B and C specify the target construction and exact training procedure.

Two manipulation checks, fixed in advance as prerequisites for opening the later transfer endpoint, ask whether the intervention affected the retained student. First, the synthetic-target recovery assay fits a fresh ridge readout on the 95,999-example WikiText training partition and evaluates the frozen student once on the untouched 1,999-example partition. It then averages target-coordinate R^2 with equal weight across coordinates. For the synthetic-response arm and the text-derived auxiliary-control arm separately, let Δ_s^{rec} be the seed-specific recovery difference between that arm's target-trained student and its seed-matched ordinary KD student. Target uptake, the recoverability of the training target from the retained student, passes only when

$$\min_s \Delta_s^{\mathrm{rec}} > 0 \qquad \text{and} \qquad L_{0.95}\left(\overline{\Delta}^{\mathrm{rec}}\right) > 0, \tag{11}$$

where $L_{0.95}$ is the lower endpoint of the two-sided t interval computed across technical seeds. Equation 11 requires the target-trained student to improve recovery in every seed and requires the mean improvement to remain positive after technical variation is included.

Second, the retained-student movement audit compares each trained student with its ordinary KD counterpart through parameter displacement, centered relative-Frobenius distance, and layer-6 centered kernel alignment (CKA) [30]. The movement rule below uses the Frobenius and CKA distances; global parameter displacement is reported separately. Let $D_{F,s}$ and $D_{\mathrm{CKA},s}$ denote the centered relative-Frobenius and linear-CKA distances from the seed-matched ordinary KD student, and let η_{CKA} denote duplicate-extraction noise, the CKA variation between repeated extractions of the same saved student. Retained-student movement passes only when every seed satisfies

$$D_{F,s} > 10^{-4} \qquad \text{and} \qquad D_{\mathrm{CKA},s} > \max\left(10^{-6}, 10\eta_{\mathrm{CKA}}\right). \tag{12}$$

Equation 12 requires movement to exceed both a fixed numerical floor and the measured extraction-noise floor in every saved seed. These measurements test retained change; they cannot identify which property of the auxiliary target caused it. The six seeds quantify technical variation rather than biological uncertainty.

Brain-response-specific attribution additionally requires an adequate text-derived comparator. The comparator audit groups the matched properties of Section 2.4 by the source of a possible alternative explanation. Training parity covers the shared optimization conditions and the early gradient magnitude, which was not retained during training. Target parity covers covariance geometry, effective rank, scale, row order, and nuisance predictability. Baseline learnability covers ordinary KD recoverability and remaining headroom; Appendices B and F list the individual checks. These training, target, and baseline properties are identification conditions because they are fixed or measured independently of the target-arm outcomes. Parameter and representation movement are post-training diagnostics that may explain how the interventions differed; they are not comparator-matching conditions. Passing every required pre-outcome or baseline axis is necessary but not sufficient for attributing an effect to brain-response content. Failure on any such axis makes the contrast non-identifying; if no required axis fails but a required measurement is missing, attribution remains unresolved. The saved block-permuted target variants trained alongside these arms are sensitivity checks and are non-identifying by construction, because they leave some rows correctly paired and can duplicate or omit a row (Appendix B). The frozen row-twin control instead applies an exact permutation and alters only the target. Because no student was trained against that control, it cannot replace the text-derived auxiliary-control arm.

The saved-student biological-transfer assay then asks whether the synthetic-response intervention improves prediction of independently recorded responses. Let $U_{ufs}^{(a)}$ be the nuisance-controlled, ROI-averaged unique R^2 for participant u, contiguous fold f, technical seed s, and training arm a. The direct biological-transfer endpoint uses the ordinary KD arm as its reference. With $F = 5$ folds and $S = 6$ seeds, define

$$\begin{aligned} d_{ufs}^{\text{direct}} &= U_{ufs}^{(\text{synthetic})} - U_{ufs}^{(\text{KD})}, \\ e_u^{\text{direct}} &= \frac{1}{FS} \sum_{f=1}^{F} \sum_{s=1}^{S} d_{ufs}^{\text{direct}}, \\ \widehat{\delta}_{\text{direct}} &= \frac{1}{9} \sum_{u=1}^{9} e_u^{\text{direct}}. \end{aligned} \tag{13}$$

Equation 13 first compares paired synthetic-response and ordinary KD students within each fold–seed cell, then averages the fold–seed cells within each participant. The final mean therefore describes biological transfer across the nine tested participants, with participants rather than folds or seeds as the biological inference units. We fixed Layer 7 as the primary biological-transfer layer before participant responses were scored and did not allow best-layer selection. We retained Layer 6, where the auxiliary target attached during training, as a prespecified mechanistic sensitivity analysis. The direct contrast answers whether the synthetic-response intervention improves recorded-response predictivity relative to ordinary KD, without attributing any difference to brain-response content.

The attribution endpoint, fixed before participant scoring, applies the same cell scores and compares each arm's gain over ordinary KD. Because the two arms share a byte-identical KD baseline in each seed, this reduces to the synthetic-response minus text-derived difference. Its interpretation remains conditional on comparator adequacy.

The same saved students support two further analyses. The saved-student target-retention analysis asks whether the synthetic target remains recoverable on held-out Tuckute sentences beyond nuisance, contrasting synthetic-response and ordinary KD students within each seed (Appendix D). The composed-path diagnostic asks how much recorded-response predictivity passes through the recovered target. Let $A_{H \to T|N}$ denote the representation-specific coefficient contribution in the train-fold model that predicts the synthetic target from nuisance features N and a frozen student representation H. Let $A_{T \to Y|N}$ denote the target-specific coefficient contribution in the train-fold model that predicts recorded responses Y from N and the target T, and let $\widehat{Y}_N$ be the nuisance-only response prediction. The composed prediction is

$$\widehat{Y}_{\text{comp}} = \widehat{Y}_N + A_{T \to Y|N}\big(A_{H \to T|N}(H_{\text{test}})\big) . \tag{14}$$

The composed score is the held-out unique R^2 of this prediction beyond nuisance, computed per participant; all fold-dependent steps are fitted on training folds (Appendix D). Comparing aligned and frozen row-twin targets tests whether sentence-level target pairing matters, while comparing synthetic-response and ordinary KD students tests whether training changed access to that path. This composed score measures predictive overlap; it does not establish causal mediation. The direct saved-student endpoint remains the primary biological-transfer endpoint.

The synthetic-response contrasts are eligible for interpretation only when perplexity and bits per byte keep the compared students within the declared quality margins of Section 3.5.

3.5 Cross-cutting estimands and decision rules

Figure 5 shows the common order from a matched analysis cell to the declared inference unit. Its three tracks separate biological inference across participants, fixed-stimulus inference across shared folds or stories, and technical inference across seeds. Focal and comparator conditions are first evaluated on the same endpoint within the smallest matched analysis cell, such as a single fold–seed cell. Technical repetitions are then combined before uncertainty is calculated across participants or fixed stimulus blocks; when technical seeds are the inference unit, each seed enters the uncertainty calculation

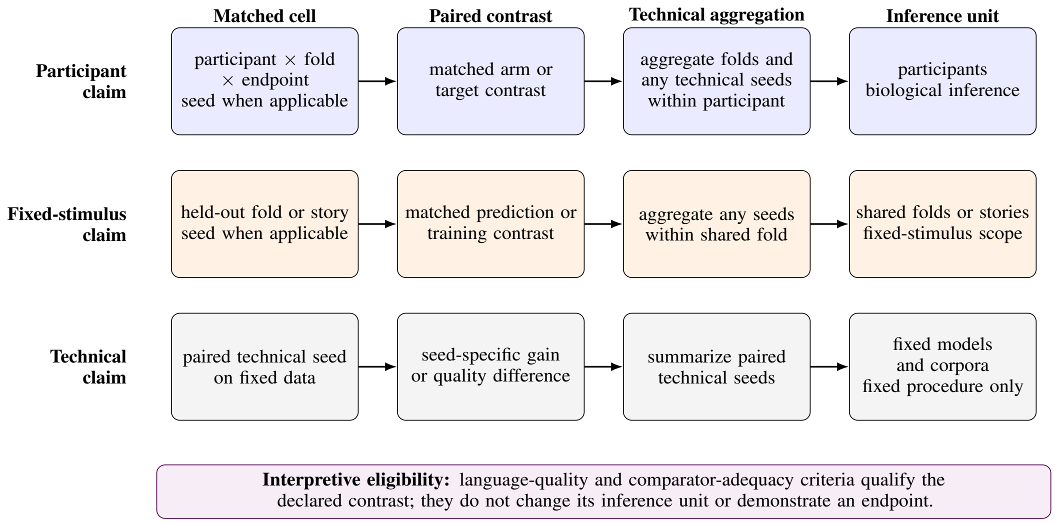


Figure 5: Aggregation paths for participant, fixed-stimulus, and technical claims.

directly. Participant estimands receive a t interval and an exact sign test; matched seed contrasts receive a descriptive t interval and, where declared, an exact sign-flip test; the target-projection, target-retention, and composed-path quantities receive Bonferroni simultaneous bounds within each declared family. Appendix D gives the exact nesting, estimators, and multiplicity rules.

A training contrast becomes eligible for interpretation only under its declared LM-quality rule. The quality metric follows the declared assay. The scaled synthetic-target family comprises all target arms of the synthetic-response route trained on the 95,999-sentence corpus; these arms and the recorded-response arms use perplexity under a shared tokenizer and scoring protocol. Cross-tokenizer or cross-family comparisons require bits per byte, and the saved-student transfer assay uses its predeclared bits-per-byte criterion even though its models share a tokenizer. Let $\mathrm{PPL}_{a,s}$ be the held-out perplexity of target arm a in seed s, and let b denote the arm against which a is compared. Let $L_{a,s}$ and $B_{a,s}$ be its total negative log-likelihood in bits and the number of decoded UTF-8 bytes in the same scored span. Let $\epsilon_{\mathrm{ppl}} = 0.05$ and $\epsilon_{\mathrm{bpb}} = 0.005$ be the declared relative-perplexity and bits-per-byte tolerances, respectively. The two principal quality criteria are

$$
\begin{aligned}
q_{\mathrm{ppl}}(a,s) &= \frac{|\mathrm{PPL}_{a,s} - \mathrm{PPL}_{\mathrm{KD},s}|}{\mathrm{PPL}_{\mathrm{KD},s}} \le \epsilon_{\mathrm{ppl}}, \\
q_{\mathrm{bpb}}(a,b,s) &= \left| \frac{L_{a,s}}{B_{a,s}} - \frac{L_{b,s}}{B_{b,s}} \right| \le \epsilon_{\mathrm{bpb}}.
\end{aligned} \tag{15}
$$

The first rule requires each scaled synthetic-target arm to remain within 5% of its seed-matched ordinary KD reference on the fixed WikiText quality probe. The second requires the saved-student comparisons to remain within 0.005 bits per byte in every seed. Passing a rule makes the contrast eligible under that metric and tolerance; it does not prove identical language behavior or remove possible differences inside the accepted margin. The paired recorded-response arm and permuted-response control are evaluated on the same fixed WikiText probe under a shared scoring implementation, with comparable perplexity by construction of the paired design. Recorded-response comparisons with ordinary KD additionally require perplexity matching across checkpoints or training curves. Without the required match, the corresponding biological contrast remains descriptive. The eligibility check qualifies interpretation but neither changes the inference unit nor supplies endpoint evidence.

minimum detectable effects (MDEs) and continuation thresholds answer narrower questions than statistical intervals. An MDE, the effect size an assay could reliably resolve under its variance model, does not imply that smaller effects are zero or biologically irrelevant. The participant analyses calculate MDEs from between-participant variation, whereas the practical-utility analysis uses variation across technical seeds. The $+0.002$ unique-R^2 threshold was reused from the biological-transfer design and fixed in advance as the rule for deciding whether the target-projection assay warrants a follow-up. It is not a universal biological-importance threshold and cannot be transferred to another response scale, nuisance model, layer, or population.

Evidence status follows the claim dependencies in Figure 1. A contrast is supported when it passes its declared decision rule, non-identifying when its comparator fails a required axis, and unresolved when a required measurement is missing or its declared interval does not decide; an unmeasured stage supplies no evidence and is not a null result. The label *not demonstrated* is reserved for an identifying, adequately measured contrast that does not pass its decision rule. We apply these statuses in Section 4 by claim role rather than experiment number or execution date.

4 Results

The subsections follow the claim dependencies in Figure 1 and the scientific roles defined in Section 3. Table 3 summarizes these role-specific outcomes using *supported*, *not demonstrated*, *unresolved*, *unmeasured*, and *non-identifying comparator*.

4.1 Controlled brain predictivity under regional and naturalistic assays

Trained representations exceeded the declared nuisance baseline in the controlled regional predictivity assay. For the primary Qwen2.5-0.5B layer-12 comparison, controlled unique R^2 was $+0.036 \pm 0.010$, where $\pm$ is the descriptive standard deviation across the five stimulus folds. The recorded trained-minus-untrained difference is $+0.050$, but the executed runner built the static nuisance block separately for each model. Because the target is participant-averaged, the folds describe stimulus-block variation and do not support participant-level inference. The assay therefore supports controlled regional predictivity for this fixed participant-averaged target.

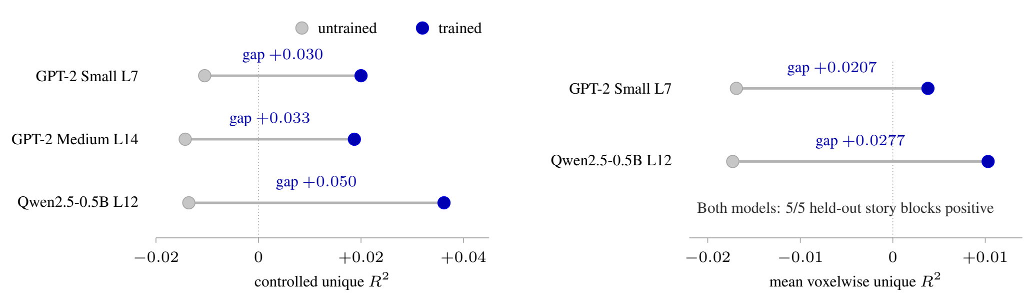
Trained representations also exceeded their matched untrained controls in the controlled naturalistic voxelwise predictivity assay when complete stories were held out. For participant UTS03, the mean differences were $+0.0207$ for GPT-2 Small and $+0.0277$ for Qwen2.5-0.5B, and all five held-out-story blocks had positive differences. Voxels are spatially dependent and story folds share training stories, so neither provides independent biological replication. This assay therefore supports a within-participant controlled brain predictivity result conditional on UTS03 and the fixed naturalistic protocol.

Together, the assays show that the tested trained LM representations improve linear prediction of held-out recorded fMRI responses beyond their model-specific nuisance features (Figure 6); learned-parameter attribution from the regional gap remains unresolved. Their estimates are not pooled because response construction, target resolution, nuisance set, and inference unit differ.

The assays do not identify a shared computation between the LMs and the brain or establish population-general controlled brain predictivity.

4.2 Observed distillation headroom and its association with language quality

The from-scratch GPT-2 comparison shows an observed teacher-relative gap, not the quality-aware headroom the intervention routes need, which is a teacher–student gap measured at comparable language quality. At the fixed Tuckute layers, the randomly initialized GPT-2 Small logit-KD student is $+0.0181$ below the GPT-2 Medium teacher in controlled brain predictivity (bootstrap probability that the student does not drop below the teacher, $p = 0.000$, over the cold arm's folds



(a) Controlled regional predictivity. **(b)** Controlled naturalistic voxelwise predictivity.

Figure 6: Assay-specific trained–untrained contrasts; scales and inference units remain separate.

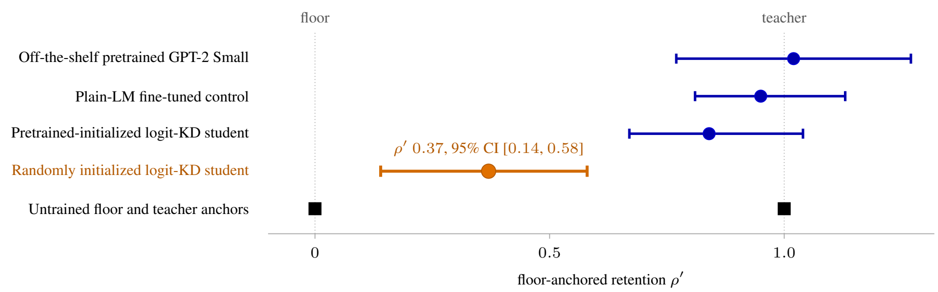
Table 3: Role-specific evidence status and decisive observations.

Scientific role	Status	Decisive observation
Controlled regional predictivity assay	**Supported**; attribution of the gap to learned parameters **unresolved**	Trained Qwen2.5-0.5B controlled unique R^2 is $+0.036 \pm 0.010$. The $+0.050$ trained-minus-untrained gap was computed with model-specific static nuisance blocks.
Controlled naturalistic voxelwise predictivity assay	**Supported**	GPT-2 Small and Qwen2.5-0.5B gaps over matched untrained controls in participant UTS03 are $+0.0207$ and $+0.0277$, positive on all five held-out-story blocks.
Distillation-headroom analysis	Observed cold-start gap **supported**; quality-aware GPT-2 warm route **unresolved**; Qwen route **unmeasured**	The cold-start GPT-2 Small gap is $+0.0181$, but initialization and LM quality differ. The warm-start comparison provides context for this route but does not establish a teacher–student gap at comparable language quality.
Recorded-response intervention, participant-averaged estimand	**Not demonstrated**	Paired gain is $+0.0081$ (five-fold t interval $[-0.0030, +0.0193]$); one fold dominates the mean.
Recorded-response intervention, participant-specific estimand	**Not demonstrated**	Nine-participant mean is $+0.00010$ (participant t interval $[-0.00037, +0.00058]$).
Target-projection assay	**Not demonstrated**; row-pairing specificity **unresolved**	Aligned-target increment is -0.004494 (two-sided 95% t interval $[-0.010695, +0.001707]$); row-twin contrast unresolved.
Synthetic-target recovery assay	**Supported**	WikiText recovery gain over seed-matched KD is $+0.078298$ (two-sided 95% t interval across technical seeds $[+0.077198, +0.079398]$).
Retained-student movement audit	**Supported**	All seeds pass; mean relative-Frobenius movement from KD is 0.2151, at acceptable LM quality.
Saved-student target-retention analysis	**Unresolved**	Fresh held-out Tuckute retention increment is -0.000855 (two-sided 95% t interval $[-0.002246, +0.000537]$).
Comparator-adequacy audit and attribution assessment	**Non-identifying comparator**	The text-derived auxiliary-control arm fails $16/37$ identification checks; post-training movement remains diagnostic and does not determine comparator validity.
Saved-student biological-transfer assay	**Not demonstrated**	Participant-level synthetic-response-minus-KD mean is -0.000014 (95% interval $[-0.000739, +0.000711]$).
Composed-path diagnostic	**Unresolved**	Synthetic-response-minus-KD increment is $+0.000031$ (two-sided 95% t interval $[-0.000080, +0.000142]$).
Practical-utility evaluation	Endpoint **unresolved**; brain-specific utility **not demonstrated**	Pereira and LeBel means are -0.019 and -0.068, both favoring the recorded-response arm; no paired endpoint interval or equivalence analysis was retained.

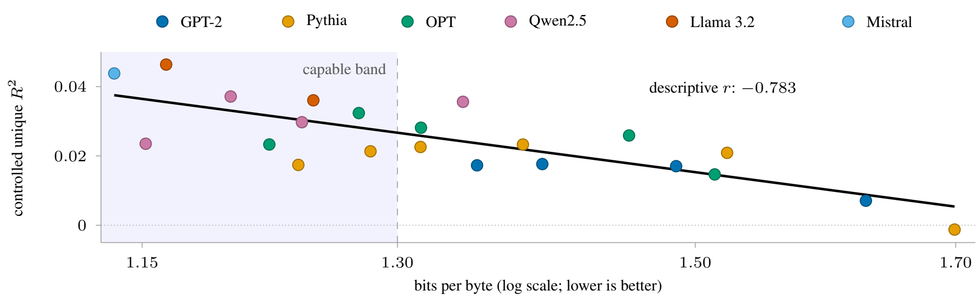
Status key. **Supported** means the declared criterion passed. **Not demonstrated** means the tested contrast did not support the claim. **Unresolved** means a required measurement is missing, or the declared interval does not decide. **Unmeasured** means the route-matched comparison was not run. **Non-identifying comparator** means the available control cannot attribute an effect to brain-response content.

and seeds). Relative to the architecture-matched untrained GPT-2 Small floor, the student retains 0.37 of the teacher-to-floor interval (Figure 7a, Equation 4). The recorded 95% bootstrap interval over folds and seeds is $[0.14,\ 0.58]$.

Language quality is a material competing explanation. The randomly initialized student remains substantially weaker as an LM, with perplexity 477, compared with 74 for the teacher. The conditions also differ in initialization and training history, not only in objective, so the estimates do not show



(a) Floor-anchored retention for the GPT-2 teacher, floor, and student conditions.



(b) Controlled brain predictivity across 22 decoder-only LMs.

Figure 7: Observed headroom and its association with language quality.

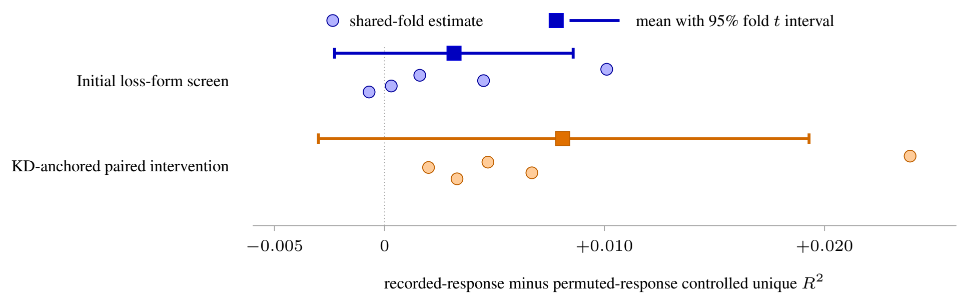
that compression or KD caused the gap. The pretrained GPT-2 Small condition, the warm route, is closer to the teacher in initialization and language quality and is the relevant reference for the synthetic-response route. The recorded-response intervention uses Qwen2.5-1.5B and Qwen2.5-0.5B, for which we did not measure route-matched teacher–student headroom.

Across 22 decoder-only LMs from six families, participant-averaged controlled brain predictivity is negatively associated with log bits per byte, where higher values indicate poorer language quality. Pearson r is -0.783, with a family-cluster 95% confidence interval (CI) of $[-0.911, -0.500]$. Within the capable-model band, the $n = 10$ models with bits per byte below 1.30, the association stays moderately negative but its interval includes zero (Figure 7b, Appendix F). Model family, scale, training data, and language quality remain entangled, so the association is a design constraint, not a causal explanation.

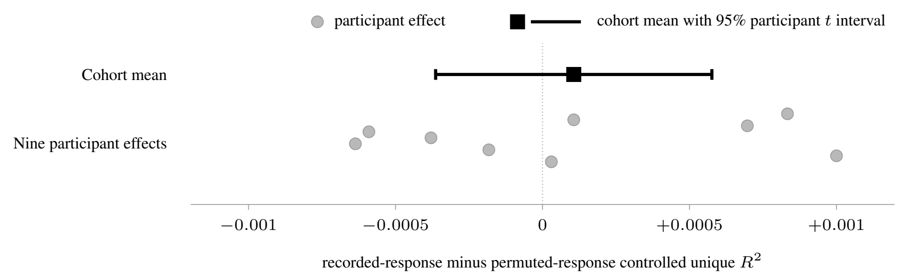
Together, the results support an observed, assay-conditional gap between the GPT-2 teacher and randomly initialized student. Quality-aware headroom remains unresolved for the warm GPT-2 route and unmeasured for the Qwen route. These results do not establish that the headroom is brain-specific or recoverable through a brain-response objective. A valid intervention comparison must therefore test the brain-response objective against a same-procedure control while maintaining comparable LM quality.

4.3 Recorded-response intervention at valid inference units

Participant-averaged recorded responses produce positive recorded-response-minus-permuted-response point estimates at the shared-fold inference unit. In the loss-form screen that selected the auxiliary response loss, the mean contrast is $+0.00316$, with a five-fold t interval of $[-0.00227, +0.00858]$. In the KD-anchored paired intervention, the mean contrast is $+0.0081$, with a five-fold t interval of $[-0.0030, +0.0193]$. The recorded-response arm and permuted-response control have perplexities 55.4 and 55.3, respectively, showing no gross quality imbalance without establishing a formal equivalence margin. Both intervals include zero, and both estimates are sensitive to a dominant fold. A pairing advantage is not demonstrated (Figure 8a).



(a) Positive averaged-target trends remain uncertain at the shared-fold unit.



(b) Participant-specific effects center near zero at the biological inference unit.

Figure 8: Recorded-response contrasts at their valid fold and participant inference units.

Averaging responses before training creates one shared stimulus-locked target, so the fold-level contrast asks whether correct stimulus–response pairing helps for that fixed target. It does not estimate the mean intervention effect across participants.

The participant-count diagnostic, re-run with random subsets so that count no longer varies with participant identity, shows no stable monotone relationship between participant count and the contrast. With model representations held fixed, averaging increases controlled brain predictivity by reducing response noise and emphasizing the shared response component; individual responses still carry measurable signal, so averaging strengthens measurement of the shared response without creating biological replication.

The participant-specific intervention does not demonstrate a participant-general advantage at the biological inference unit, the participant. Aggregating seed and fold repetitions within each participant (Section 3.3), the mean recorded-response-minus-permuted-response effect across nine participants is $+0.00010$, with participant t interval $[-0.00037, +0.00058]$. The exact sign test over the same participants gives 5 of 9 positive, $p = 0.50$. The shared-fold summary is also centered near zero at $+0.00026$, with five-fold t interval $[-0.0004, +0.0009]$ (Figure 8b).

The minimum detectable effect at 80% power, computed from the observed variance, is approximately 0.0006–0.0013 for the participant analysis. This range is far below the earlier participant-averaged point estimate of $+0.0081$. The participant analysis would therefore have detected an individual-level effect of that magnitude with at least 80% probability, although the two estimates belong to different response constructions and estimands. The participant result does not prove an exactly zero effect.

The tested variants address specific alternative explanations, but none provides evidence that changes the participant-general conclusion (Table 4). Appendices C, E, and F give objectives and schedules, variant dispositions, and full numerical tables.

Across the tested recorded-response interventions, a participant-general advantage of correct stimulus–response pairing is not demonstrated for the tested targets, models, objectives, cohorts, and readouts; this is not a universal null. The synthetic-response route is separate, and Section 4.4 first evaluates its declared 50-component linear pathway from the synthetic target to recorded responses.

Table 4: Recorded-response robustness checks and their scoped conclusions.

Variant	Strongest valid observation	Scoped conclusion
Frozen auxiliary head	The frozen-head arm does not exceed its own permuted control.	A fixed readout produces no pairing-specific gain.
Higher-capacity LoRA	Participant mean $+0.00043$, with t interval $[-0.0005, +0.0013]$; the shared-fold reading agrees.	Additional adapter capacity at the fixed loss weight does not rescue the effect and does not produce a stronger manipulation.
Contrastive objective	Participant mean -0.00019, with t interval $[-0.0008, +0.0005]$; the shared-fold reading agrees.	Changing the loss does not rescue the effect at five-ROI granularity.
Naturalistic voxelwise LoRA	The recorded-response arm does not produce the required improvement over its starting model and permuted control.	The intended manipulation is not demonstrated; this is not a naturalistic population null.
Naturalistic full fine-tuning	Lower-intensity training shows no brain-specific paired gain across three participant probes; higher-intensity training collapses previously learned language-model quality.	The tested full-parameter route does not rescue the effect, and no population-general inference is supported.

4.4 Measurability of the 50-component linear target projection

Section 3.2 defines the aligned-target increment $A(u)$ over nuisance and the correct-pairing increment $Q(u)$ over the frozen row-twin control, both aggregated across folds so that the participant is the biological inference unit.

The aligned-target increment does not pass the continuation rule carried over from the biological-transfer design. The rule was fixed in advance and requires the simultaneous family-95% one-sided upper bound to exceed τ_{cont}. Let $\bar{A}$ be the participant mean of $A(u)$ ($n = 9$). Let $U_{95\%}^{\mathrm{family}}(\bar{A})$ be its simultaneous family-95% one-sided upper bound and τ_{cont} the continuation threshold. The result is

$$\begin{aligned} \bar{A} &= -0.004494, \\ \mathrm{CI}_{95\%}(\bar{A}) &= [-0.010695, +0.001707], \\ U_{95\%}^{\mathrm{family}}(\bar{A}) &= +0.001707 < \tau_{\mathrm{cont}} = +0.002. \end{aligned} \tag{16}$$

Only 3 of 9 participant effects are positive, with exact sign test $p = 0.5078$, and every leave-one-out mean, dropping any single participant or stimulus block, remains negative. The bound excludes a participant-mean effect of at least τ_{cont} without showing the effect to be zero, and it does so because the point estimate is negative rather than because the estimate is precise. The observed-variance 80% MDE for this contrast is 0.008608.

The participant mean of the correct-pairing increment $Q(u)$ is $+0.001758$. Its two-sided 95% t interval is $[-0.003575, +0.007091]$. 5 of 9 participant contrasts are positive, with exact sign test $p = 1.0$, but the interval spans zero (Figure 9).

On the fixed sentence set, linear measurability of the target projection is not demonstrated, and row-pairing specificity remains unresolved. Training used the full 20,484-coordinate target, so not clearing the continuation rule says nothing about full-dimensional or nonlinear access to that target. The assay leaves open total predictivity, predictivity redundant with the nuisance model, and effects under another target transform, cohort, or endpoint. The next subsection asks independently whether training produced target uptake and retained-student movement, and whether the available comparator attributes any arm difference to brain-response content.

4.5 Synthetic-response manipulation checks and attribution status

Training changed the saved student. Across the $n = 6$ saved technical seeds, a fresh readout recovers more of the synthetic target from the synthetic-response student than from its seed-matched ordinary KD student, by $+0.078298$ under Equation 11, with two-sided 95% seed-axis t interval

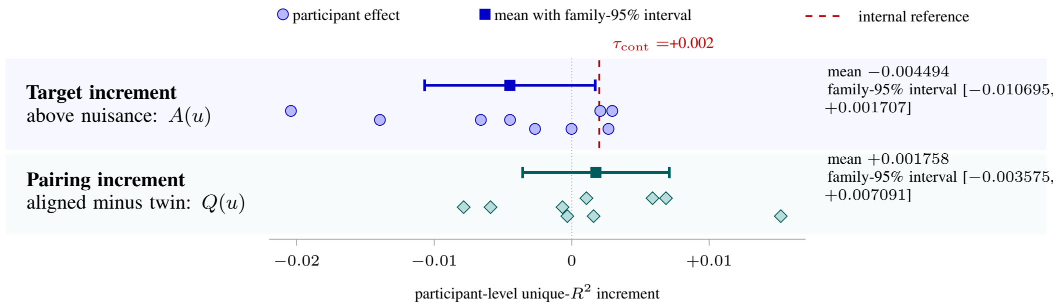


Figure 9: Participant means and two-sided 95% t intervals for the aligned-target increment and the row-pairing contrast; the threshold guide applies to the increment only.

[+0.077198, +0.079398]. Every saved technical seed improves recovery, so the saved procedure passes the WikiText target-uptake check and uptake is supported. These seeds quantify technical stability, not biological or model-population uncertainty.

The retained student also moved away from its ordinary KD counterpart, by a mean layer-6 centered relative Frobenius distance of 0.2151. Every saved technical seed passes the movement check. Across the direct saved-arm comparisons, the largest absolute language-quality difference is 0.002579 bits per byte, below the frozen 0.005 margin. The movement is therefore established at language quality within the declared margin, but not shown to arise from brain-response content.

On fresh held-out Tuckute sentences, scored beyond the nuisance model, that recovery gain over ordinary KD does not reappear. Absolute target extractability from the synthetic-response students is +0.040725, with two-sided 95% t interval [+0.039053, +0.042398]. The incremental synthetic-response-minus-ordinary KD estimate is instead −0.000855, with two-sided 95% t interval [−0.002246, +0.000537]. Thus, the target is linearly recoverable from the trained students on these sentences, but whether synthetic-response training increased its recoverability beyond ordinary KD remains unresolved (Figure 10a).

Differences between the synthetic-response and text-derived arms in these manipulation results cannot be attributed to brain-response content, because the text-derived auxiliary-control arm fails 16/37 identification checks (Figure 10b). Baseline learnability differs most. Before intervention, ordinary KD representations recover mean target R^2 of 0.595327 versus 0.911687 for the synthetic and text-derived targets, leaving 4.582 times as much remaining room for the synthetic target. The arm also fails the required target-parity margins for covariance geometry, effective rank, and low-level nuisance predictability. Training parity is unresolved for initial auxiliary loss and gradient magnitude, which were not retained.

Separately, 14/18 post-training diagnostic margins fall outside the tolerances originally set for them, which were descriptive margins rather than declared decision rules. The pre-outcome and baseline mismatches can affect apparent learnability and retained-student movement without any difference in brain-response content.

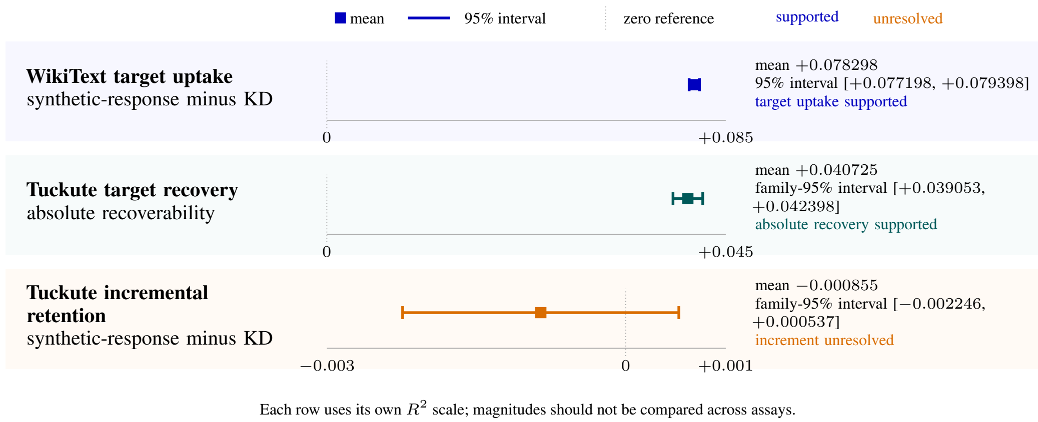
The saved block-permuted arm cannot resolve this attribution problem, because it leaves some rows correctly paired and can duplicate or omit a row (Section 3).

The intervention therefore produced target uptake and retained-student movement, but the available comparator remained non-identifying for attribution; the next subsection tests biological transfer under that limit.

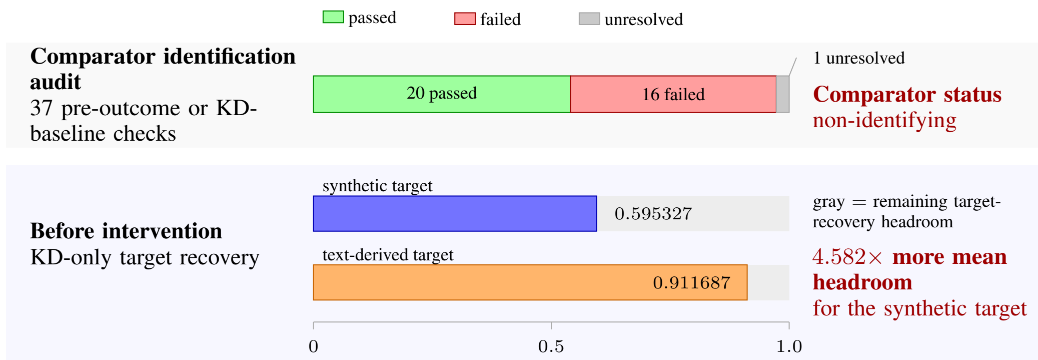
4.6 Saved-student biological-transfer assays

The direct biological-transfer assay compares each synthetic-response student with its seed-matched ordinary KD student, averaging folds and technical seeds within each participant so that the participant is the biological inference unit ($n = 9$). The participant-level result is

$$\begin{aligned} \widehat{\Delta}_{\mathrm{direct}} &= -0.000014, \\ \mathrm{CI}_{95\%} &= [-0.000739, +0.000711]. \end{aligned} \tag{17}$$



(a) Manipulation and target-retention results on their assay-specific scales.



(b) Comparator-audit outcomes and baseline target-recovery mismatch.

Figure 10: Synthetic-response manipulation and attribution evidence.

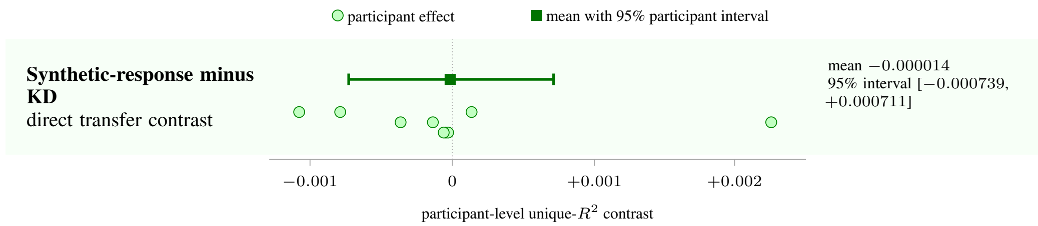
Here $\widehat{\Delta}_{\mathtt{direct}}$ is the participant-mean change in held-out unique R^2 from synthetic-response training relative to ordinary KD (Figure 11a). The participant t interval spans zero, so participant-general biological transfer is not demonstrated in the tested cohort. Its scope remains limited to the fixed target, student, layer, nuisance design, sentences, participants, and ROI endpoint.

The attribution endpoint, each arm's gain over ordinary KD, is numerically positive.

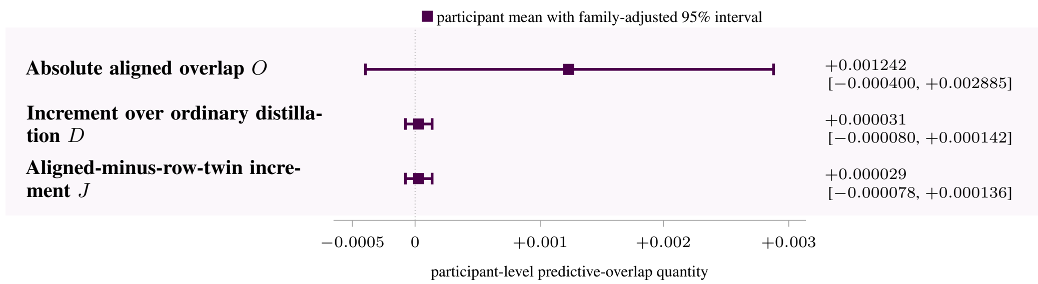
$$\begin{aligned} \widehat{\Delta}_{\mathtt{synthetic-text}} &= -0.000014 - (-0.000150) = +0.000136, \\ \mathrm{CI}_{95\%} &= [-0.000505, +0.000777]. \end{aligned} \tag{18}$$

The positive relative contrast arises because the text-derived auxiliary-control arm is more negative than ordinary KD, not because the synthetic-response arm has a positive direct effect. The participant t interval also spans zero, and 4 of 9 participant contrasts are positive with exact sign test $p = 1.0$. The one-sided 95% participant upper bound $+0.000653$ lies below the continuation threshold $+0.002$, so a participant-mean effect of at least that size is excluded, though not shown to be exactly zero. The observed-variance 80% MDE for this contrast is 0.000890. Because the text-derived auxiliary-control arm is not adequately matched (Section 4.5), this contrast cannot identify an effect of brain-response content.

The composed-path diagnostic instead measures whether the synthetic-target component that a fresh readout recovers from the frozen student also predicts recorded responses, through three



(a) Direct synthetic-response-versus-ordinary KD effects across participants.



(b) Composed-path diagnostic two-sided 95% t intervals.

Figure 11: Direct biological transfer and the composed-path diagnostic.

participant-level estimands (Figure 11b).

$$\widehat{O} = +0.001242, \qquad \mathrm{CI}_{95\%}(\widehat{O}) = [-0.000400, +0.002885], \tag{19}$$

$$\widehat{D} = +0.000031, \qquad \mathrm{CI}_{95\%}(\widehat{D}) = [-0.000080, +0.000142], \tag{20}$$

$$\widehat{J} = +0.000029, \qquad \mathrm{CI}_{95\%}(\widehat{J}) = [-0.000078, +0.000136]. \tag{21}$$

Here, O measures absolute aligned predictive overlap, D measures its increment over seed-matched ordinary KD, and J asks whether that increment is larger for correctly aligned targets than for the frozen row twin. Every two-sided 95% t interval spans zero. All three quantities therefore remain unresolved under participant inference.

The direct assay does not demonstrate participant-general biological transfer, and the composed-path diagnostic leaves its overlap quantities unresolved.

4.7 Practical-utility evaluation

The only completed practical-utility evaluation compares the averaged recorded-response arm with its approximately quality-matched permuted-response control across $n = 8$ technical training seeds. Negative values of that contrast (Equation 9) favor the recorded-response arm. Averaging that contrast across seeds gives

$$\overline{\Delta R}_{\mathrm{Pereira}} = -0.019, \qquad \overline{\Delta R}_{\mathrm{LeBel}} = -0.068. \tag{22}$$

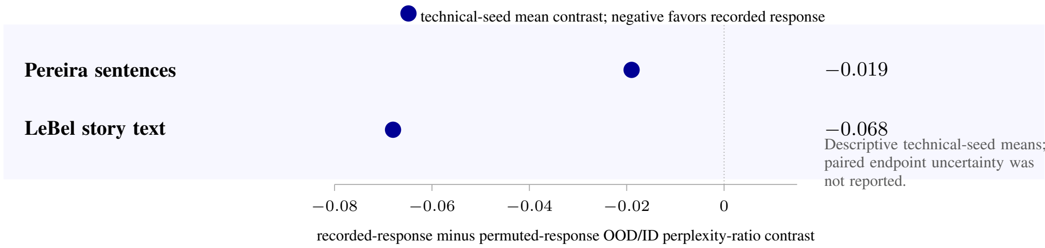
Both mean contrasts, for Pereira sentences [28] and LeBel story text [25], favor the recorded-response arm (Figure 12a). The completed analysis retained no paired technical-seed uncertainty interval or formal equivalence test. The endpoint contrasts are therefore unresolved.

The endpoint means can be credited to brain-response training only if that training reliably changed controlled brain predictivity, the prerequisite contrast A_s of Equation 10. Its seed-level summary is

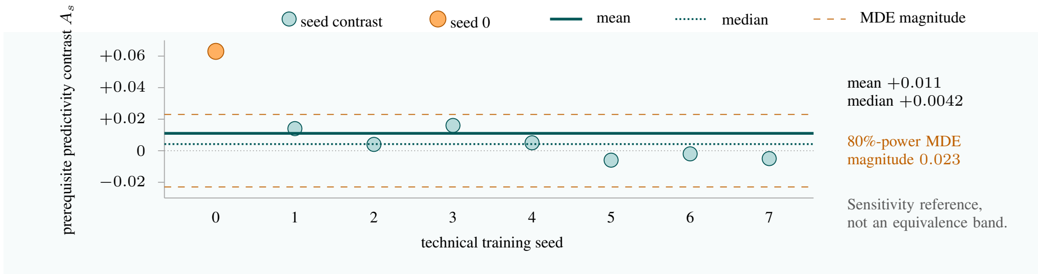
$$\mathrm{mean}_s(A_s) = +0.011, \qquad \mathrm{median}_s(A_s) = +0.0042, \qquad \mathrm{MDE}_{80\%} = 0.023. \tag{23}$$

The MDE is a sensitivity reference, not an acceptance threshold or equivalence margin.

Only $5/8$ of the technical-seed prerequisite contrasts are positive. Seed 0 contributes $+0.063$, whereas the remaining seeds average approximately $+0.004$. The mixed signs and seed-0 influence leave the controlled-predictivity prerequisite not demonstrated across technical seeds (Figure 12b).



(a) Recorded-response-minus-permuted-response utility contrasts.



(b) Technical-seed prerequisite contrasts with an observed-MDE sensitivity reference.

Figure 12: Descriptive endpoint contrasts and their prerequisite contrast in controlled brain predictivity.

The completed analysis does not demonstrate a reliable brain-response-specific practical advantage for these corpora, this student–teacher pair, and this recorded-response procedure at approximately matched LM quality. The result also does not establish the absence of an endpoint advantage. Supporting reproductions and stopped routes remain in Appendix E.

5 Discussion

In the systems tested, the experiments support two outcomes, controlled brain predictivity and a student that synthetic-response training measurably changed, but not a brain-specific training advantage. Controlled brain predictivity establishes that trained LM representations contain information useful for predicting recorded fMRI responses. Post-training target uptake and retained-student movement, both supported, establish that the synthetic-response objective altered the saved student. These results do not by themselves establish brain-response-specific attribution, biological transfer, or practical utility. Support for the recorded-response and synthetic-response routes stops at different stages, in both cases before a participant-general training benefit is established.

5.1 Where support is lost across the two intervention routes

Controlled brain predictivity supports the measurement claim of the recorded-response route. Under the declared controls, model representations do predict recorded responses. The recorded-response intervention does not demonstrate a participant-general pairing advantage. Training with participant-averaged responses produced positive trends for a fixed shared target. Those comparisons did not demonstrate a reliable pairing advantage even at the unit they can address, a held-out fold of the one shared averaged target rather than an individual participant. Averaging changes the estimand and does not create participant replicates. When responses and technical repetitions were aggregated within participants, correct stimulus–response pairing did not demonstrate a participant-general advantage. Greater adapter rank did not produce a stronger manipulation, and the contrastive, naturalistic, and full-parameter analyses bound their conclusions to the loss, data, and optimization regimes they tested. These variants do not establish that every recorded target or training procedure would fail. The practical-endpoint means favor the recorded-response arm, but with no paired uncertainty and an unstable controlled-predictivity prerequisite the endpoint is unresolved and cannot be read as the payoff of a reliably induced brain-response-specific change.

The synthetic-response evidence separates into parallel claims rather than one failure sequence. The unique linear predictivity of the 50-component PCA target representation did not meet the continuation rule, while its advantage over the frozen row-twin control, the same target rows with their sentence pairing broken, remained unresolved. Post-training target uptake and retained-student movement show that training altered the saved student while measured LM quality remained within the declared margin. They do not change the earlier measurability finding. The text-derived auxiliary-control arm is non-identifying, the direct participant-level assay does not demonstrate biological transfer, and the composed-path diagnostic remains unresolved. A supported manipulation can therefore coexist with biological transfer that is not demonstrated.

The nuisance and untrained-network controls make it unlikely that no controlled signal was present at all, and the manipulation and quality checks make it unlikely that training either failed to alter the student or badly damaged its language quality. Together, the controls narrow the explanation without identifying it. Several possibilities remain compatible with the evidence, namely that the targets may not carry the needed information; that it may be present but out of reach of the declared linear readout; that the optimization, model, or data scale may be too small to use it; or that it may be participant-specific. These possibilities are hypotheses suggested by where support was lost, not causes established by the experiments.

5.2 Compatibility with positive prior studies

The route-specific account of where support stopped does not directly contradict positive brain-response-guided studies. Prior work differs from this study in modality, supervision, training budget, and endpoint (Section 2.3), and those differences change both the intervention and the effect being estimated. The present results are conditional on the tested student, targets, controls, and inference units. Because the designs are not equivalent, the results here cannot directly adjudicate those studies. No single design difference is shown to have caused the divergence.

What a positive contrast establishes depends on its reference condition (Section 2.4). Improvement over a pretrained or language-only model leaves the effect of the brain-response objective entangled with generic quality differences, while the ordinary KD arm isolates that effect, and comparable LM quality limits those differences. Comparator and endpoint choice bound the reading the same way

(Section 2.4), since a control that is not route-appropriate cannot attribute a gain to brain-response content, and recovery of the optimized target is not the biological transfer or task gain a different endpoint would test. Training seeds can establish procedural stability, but they cannot replace participant-level uncertainty when the claim concerns people.

We keep measurement and manipulation distinct from attribution, transfer, and utility, and specify the comparator, quality condition, endpoint, and inference unit needed for each claim. Whether materially different prior results would pass those additional stages remains an open empirical question.

5.3 Design implications for future brain-response-guided training

The two routes lost support at different points, so they call for different design changes. For recorded-response training, response construction and inference must match the intended claim. A target averaged across participants estimates the shared response. Participant-general claims instead require population inference that keeps the participant as the biological unit. The two estimands are complementary, not interchangeable.

For synthetic-response training, the point at which support stopped is a reason to move several checks before expensive optimization. Here the target-projection assay and the comparator audit were run after training, when neither could redirect it. The training target, after compression to 50 principal components, should first be evaluated on held-out data from the intended recorded responses. That assay should use the planned nuisance model and readout, a frozen row-twin control, the inference unit required by the claim, and a prespecified assay-specific continuation rule, because the executed assay inherited its threshold from the biological-transfer design rather than setting one for the linear pathway it tested. Content attribution requires a separately trained text-derived auxiliary-control arm. Before training, that auxiliary target should be checked for comparable scale, covariance geometry, nuisance predictability, baseline recoverability, remaining headroom, and held-out learnability. Initial auxiliary losses and gradient scales should also be retained so that optimization pressure can be audited directly. Passing these checks would establish a better-controlled opportunity under the declared assay, not attribution or transfer. Not meeting the 50-component PCA linear criterion would still not rule out every nonlinear intervention.

Once target uptake and retained-student movement are supported at acceptable LM quality, evaluation must move beyond the optimized target. Attribution requires comparison with the text-derived auxiliary-control arm once those pre-training checks have passed. Biological transfer then requires an independent recorded-response endpoint, with technical repetitions aggregated within each participant before population inference. These outcomes must remain separate. Utility can be read as the payoff of improved controlled brain predictivity only once that prerequisite is reliably established and the practical endpoint is reported with paired uncertainty. The completed utility evaluation belongs to the recorded-response route and does not estimate synthetic-route utility.

The stage status should determine the next use of data and compute. A supported stage permits the next test but does not license its claim. A stage at which the claim is not demonstrated justifies further work only when additional participants, a more informative endpoint, or a redesigned assay could change the decision. An unresolved stage calls for a targeted test that can distinguish the remaining alternatives. A non-identifying comparator requires redesign before attribution can proceed. When no feasible follow-up can change the finding, further dependent evidence has little interpretive value and the route should stop or redesign its target, control, or assay. Two further design changes follow from where this study lost support. Separating the effect of a training objective from language quality requires a model set in which alignment varies while quality stays approximately flat, because only there can an alignment difference be attributed to the objective rather than to quality. Testing brain-response content against an independent recording modality requires both a phase-randomized response control and a geometry-matched control, since a phase-randomized twin has already reproduced an apparent cognition-specific advantage in a simpler target (Appendix E). These recommendations follow from the diagnostic pattern observed here, not from an experimentally proven universal procedure. They apply within the populations, measurements, and intervention spaces set out in Section 6.

6 Limitations

Section 5 identified where the evidence stopped supporting the recorded-response and synthetic-response routes. Those conclusions are bounded by the populations and measurements covered, by what each analysis can identify, and by the range of interventions and utility endpoints searched.

6.1 Population, data, and measurement scope

The biological evidence is limited to fMRI responses to English-language stimuli from the studied cohorts. The sentence-level assays summarize responses within fixed left-hemisphere language ROIs. The naturalistic voxelwise predictivity assay uses voxels selected for response reliability in one participant who was scanned extensively while listening to stories. Some recorded-response targets average responses across participants; the participant-level analyses instead retain one within-participant summary effect per person. The conclusions therefore concern these isolated-sentence and spoken-story response summaries, not language processing across the whole brain or population at large.

The study does not cover recording modalities with finer temporal resolution, populations that differ in language background or in clinical and developmental status, non-English or nonlinguistic stimuli, or broader reading, listening, and interactive contexts. It also does not test whether nonlinear mappings could recover information that the declared linear ridge readouts miss. The route-specific conclusions therefore do not extend to these settings.

6.2 Identification and inference limitations

The nuisance set, comparator, readout, and inference unit determine what each result identifies. Unique predictivity rules out only the nuisance features that were actually included, and no other alternative explanation. In the regional assay the untrained arm's static-embedding nuisance was built from its own model, so the trained-versus-untrained contrast leaves unresolved whether the predictivity gain comes from the learned parameters. The text-derived auxiliary-control arm differs from the synthetic target in baseline predictability, remaining headroom, covariance geometry, effective rank, and nuisance predictability. Its cross-target contrast therefore cannot identify the contribution of brain-response content. The target-projection result is restricted to the declared 50-component linear pathway, and specificity against the frozen row-twin control remains unresolved. Section 3.5 defines the biological, fixed-stimulus, and technical tracks that keep the inference units apart, and even correct aggregation cannot extend a fixed or shared stimulus set to all language stimuli. Appendix D formalizes these identification and dependence boundaries.

The study contains positive checks for separate parts of the argument, with the measurement assays supporting controlled brain predictivity and the synthetic-response intervention satisfying the target-uptake and retained-student-movement criteria. It does not contain a positive control, that is, a brain-response intervention already expected to improve participant-level biological transfer through the complete evaluation path. The component checks do not establish that the full intervention-to-transfer path is sensitive enough to reveal a genuine brain-specific training advantage.

The recorded-response participant analysis (Section 4.3) reports an interval and a minimum detectable effect but declared no continuation threshold, so it bounds sensitivity without excluding any effect size. The threshold exclusions in Sections 4.4 and 4.6 belong to the target-projection assay and to the attribution endpoint, the cross-target contrast above, not to the direct biological-transfer contrast, which is bounded only by its participant interval. Smaller average contrasts across participants, heterogeneous participant effects, and effects in other cohorts remain compatible with the evidence.

6.3 Intervention and utility scope

The central synthetic-response route tested a GPT-2 Small student paired with a GPT-2 Medium teacher, representing one teacher–student scale pairing within one architecture family (Section 3.4). The recorded-response route instead distils a Qwen2.5-1.5B teacher into a Qwen2.5-0.5B student (Section 3.3). The routes run in different model families, and each is evaluated only against its own control under the eligibility rule in Section 3.5, so no result compares one route with the other and no reported contrast depends on the family difference. Across both routes, the study did not jointly explore broader student architectures, objective families, loss weights, schedules, data volumes, target constructions, or participant-personalization strategies. The naturalistic and full-parameter variants were probes of mechanism, or three-participant existence probes, not participant-general tests, so they do not broaden population inference. Section 4.3 and Appendix E report the individual variant designs and dispositions.

The practical-utility evaluation in Section 4.7 rests on a narrow set of endpoints. Perplexity served both as a language-quality control and as the primary out-of-domain utility measure; the study therefore did not evaluate task accuracy, calibration, robustness, sample efficiency, or deployment-specific value. The completed endpoint analysis reports descriptive means across technical seeds but no paired endpoint interval or equivalence test, so the direct endpoint advantage remains unresolved. The recorded-response route also did not establish the technically stable change in controlled brain predictivity that would be required to interpret that endpoint contrast as brain-specific utility. No synthetic-response utility test was run. Sections 4.4 and 4.6 report that the target-projection assay did not meet its continuation rule and that participant-level biological transfer was not demonstrated. The supporting studies summarized in Appendix E tested narrow design levers, stopped before intervention, or stopped because the planned measurement was unusable. They therefore do not broaden the set of completed endpoints that identify brain-specific utility. Whether an intervention with an identified biological effect would also help at untested practical endpoints remains open.

7 Conclusion

For the systems tested using fMRI responses to English-language stimuli, neither intervention route established a participant-general training benefit attributable to brain-response content. The recorded-response route supported controlled brain predictivity, but across the tested objectives and variants it did not demonstrate that correctly paired responses beat permuted responses for participants in general. The WikiText checks in the synthetic-response route supported target uptake (the target became easier to recover from the student) and retained-student movement (the retained student still differed from its baseline), both at acceptable measured language quality. However, the target-projection assay did not meet its continuation rule, and the text-derived auxiliary-control arm was non-identifying, too poorly matched to attribute any difference to brain-response content. Direct participant-level biological transfer over ordinary KD was also not demonstrated. The utility endpoint for recorded-response training remained unresolved, and brain-specific utility was not demonstrated. The tested routes therefore do not establish brain alignment as a usable training signal, but they do not rule out a smaller benefit in these systems or a benefit from other targets, interventions, or populations. The supported contribution is narrower than the question this thesis asked. It comprises controlled brain predictivity in the declared assays, a demonstrated ability to change the student through synthetic-response training, and an evidence framework that keeps those findings distinct from attribution, biological transfer, and utility.

A controlled brain predictivity score alone does not establish a usable training signal. Within the evidence criteria used in this thesis, target uptake and retained-student movement must pass before transfer or attribution is evaluated, and a utility claim framed as the payoff of improved controlled brain predictivity requires that the predictivity improvement itself be reliably established first. After optimization, several outcomes must be evaluated separately, namely target uptake and retained-student movement at acceptable measured language quality; direct participant-level transfer over ordinary KD; brain-response-specific attribution against an adequate route-appropriate control; and any claimed utility endpoint. For recorded-response training, a permuted-response control tests whether correct stimulus–response pairing matters. For synthetic-response training, an adequately matched text-derived auxiliary-control arm tests whether any observed benefit is specific to brain-response content. Seeds, folds, voxels, and ROIs cannot substitute for participants when the claim concerns people. Kept separate, these claim dependencies distinguish what an intervention achieved from what a usable training signal still needs.

A Evidence Records and Provenance

Every experiment reported here has a numbered record, E001 to E030, in the accompanying repository under `docs/experiments/`. Each record states what was executed, what its retained results support, and which interpretation survived review. The main text states the resulting claims. This ap-

Table 5: Scientific roles, final evidence owners, and settled conclusions.

Scientific role	Evidence record(s)	Settled conclusion	Reported in
Controlled regional predictivity assay	E002	Controlled Tuckute ROI predictivity is supported under contiguous folds, nuisance subtraction, and an architecture-matched untrained control.	4.1
Controlled naturalistic voxelwise predictivity assay	E006	Trained representations outperform matched untrained representations for UTS03 and five held-out story blocks. The assay does not support claims about individual voxels or a population of participants.	4.1
Distillation-headroom analysis	E003; E015 supports the quality analysis	The cold GPT-2 route shows an observed teacher-relative gap; quality-aware warm-route headroom is unresolved and Qwen-route headroom is unmeasured.	4.2
Recorded-response intervention, participant-averaged estimand	E004 and E005; E010 and E014 bound the averaging interpretation	Fold-unit reanalyses of E004 and E005 supersede their earlier excludes-zero readings. E010 does not identify a participant-count response. E014 shows that averaging reduces response noise and emphasizes shared response while changing the estimand.	4.3
Recorded-response intervention, participant-specific estimand	E008; E011, E013, and E017 bound tested variants	No participant-general pairing advantage is demonstrated across the tested objectives and variants.	4.3
50-component linear target-projection assay	E030	The fixed 50-component PCA target does not meet the continuation rule of $+0.002$ fixed for it in advance. Aligned-minus-row-twin specificity remains unresolved.	4.4
Synthetic-target recovery assay	E016	WikiText assays support target uptake.	4.5
Retained-student movement audit	E025; E016 supports the earlier assay	Retained-student movement is supported at acceptable measured LM quality.	4.5
Comparator-adequacy audit and attribution assessment	E026; E016 supplies the original comparator	The saved text-derived auxiliary-control arm is non-identifying because pre-outcome target and KD-baseline comparability checks do not hold; post-training movement remains diagnostic.	4.5
Saved-student target-retention analysis	E030	The exact target is recoverable from the synthetic-response students on Tuckute, while incremental retention beyond ordinary KD remains unresolved.	4.5
Composed-path diagnostic	E030	Within the declared linear pathway, predictive overlap, incremental overlap, and the aligned-over-row-twin increment remain unresolved across participants. These diagnostics do not constrain the independent direct-transfer contrast.	4.6
Saved-student biological-transfer assay	E025; E026 bounds attribution	Participant-level biological transfer is not demonstrated. The direct synthetic-response-minus-KD contrast is near zero, while the text-derived comparison is non-identifying.	4.6
Practical-utility evaluation	E009	Mean out-of-domain ratios favor the recorded-response arm, but paired endpoint uncertainty was not reported; the prerequisite contrast is unstable across technical seeds, so brain-specific utility is not demonstrated.	4.7

pendix links those claims to the experiments behind them. It names the experiment that settled each scientific role.

Not every record supports a settled scientific conclusion. Some had their reading narrowed by a recorded correction, some are limited to a local or diagnostic scope, and some stopped before an interpretable test. Each record states its own limit, and Appendix E describes the supporting analyses and stopped routes.

Table 5 follows the claim roles used in Sections 3 and 4. Each row names one scientific role, the experiment that settled it, and the Results subsection where its evidence is interpreted.

Appendices C, D, and F report the corresponding training settings, estimators and inference rules, and complete numerical and sensitivity tables.

B Data, Responses, Targets, and Controls

This appendix specifies the data and target constructions behind the assays in Sections 3 and 4. It moves from recorded-response inclusion and aggregation, through nuisance features and fold-local preprocessing, to synthetic and text-derived targets and their controls.

B.1 Data inclusion and response construction

The recorded-response analyses use two complementary datasets. The Tuckute resource provides isolated sentences and regional responses from multiple participants, whereas the LeBel resource provides continuous stories and voxelwise responses from one deeply sampled participant. The Tuckute condition-B substrate is 1,000 English sentences ordered by `item_id`, together with five left-hemisphere language ROIs, namely `AntTemp`, `IFG`, `IFGorb`, `MFG`, and `PostTemp`.

The Tuckute resource was acquired during silent sentence reading with 3-T fMRI, a 2.0-s repetition time (TR), and 2-mm isotropic voxels [24]. This work inherits the released `response_target`; it does not reconstruct voxel time series or refit the functional localizers. In the released target, voxel responses were z-scored within scanning session and then averaged within each ROI. The remaining operations select condition B, order rows by `item_id`, and construct either an averaged or participant-specific response matrix.

The participant-averaged construction combines, cell by cell, the five participants the Tuckute release designates as its training set. Their unique identifiers (UIDs) are 848, 853, 865, 875, and 876. A missing contributor is skipped, but the resulting sentence-by-ROI matrix must be complete before analysis. This response is used by the controlled regional assay and the participant-averaged recorded-response intervention. It estimates predictivity for the shared response of this fixed group because averaging occurs before readout fitting and before held-out R^2 is calculated.

Participant-specific construction retains one complete sentence-by-ROI matrix per person. UID 853 is excluded because 60 of its sentence–ROI cells are missing, leaving the five-ROI grid incomplete. The inclusion rules differ because cell-wise averaging can use whichever entries are present, whereas participant-level inference requires a complete matrix for each person; those 60 averaged cells therefore rest on four contributors instead of five. This construction distinction does not assume that the missing entries are ignorable for every estimand. Ten participants have condition-B coverage; the nine that remain after this exclusion are UIDs 797, 837, 841, 848, 856, 865, 875, 876, and 880. Within this set, $\{848, 865, 875, 876\}$ form the inherited training-participant subgroup and $\{797, 837, 841, 856, 880\}$ form the held-out-participant subgroup. Appendix D specifies the fold and seed aggregation; the nine retained participants are the biological inference units. Participant averaging changes both the signal-to-noise ratio and the estimand; it does not create additional independent participants.

The naturalistic assay uses participant UTS03 from the LeBel story-listening resource [25]. The responses were acquired at a nominal TR of 2.0 s. The present implementation places the stimulus features and released response matrices on the 2.0045-s effective sampling grid recorded in the released dataset metadata. Each released HDF5 `data` matrix contains one row per scanner sample and 95,556 cortical-voxel columns. This work begins from those matrices and does not rerun reconstruction, motion correction, or cortical registration.

The 20 assay stories follow their recorded order and are divided into five contiguous four-story folds.

Fold	Held-out stories
1	`adollshouse`, `adventuresinsayingyes`, `afatherscover`, `againstthewind`
2	`alternateithicatom`, `avatar`, `backsideofthestorm`, `becomingindian`
3	`beneaththemushroomcloud`, `birthofanation`, `bluehope`, `breakingupintheageofgoogle`
4	`buck`, `catfishingstrangerstofindmyself`, `cautioneating`, `christmas1940`
5	`cocoonoflove`, `comingofageondeathrow`, `exorcism`, `eyespy`

The separate story `wheretheressmoke` is excluded from model fitting and scoring and is used only for voxel reliability selection.

Reliability is estimated from the ten released repeats of `wheretheressmoke`. With random seed 0, each of 50 random splits assigns five repeats to each half. Responses are averaged within each half, correlated voxelwise across the 291 scanner samples, and corrected with the Spearman–Brown formula. The mean corrected correlation across the 50 splits is the voxel's reliability estimate. Voxels with reliability strictly greater than 0.5 are retained, leaving 11,442 voxels. Because the repeated story is independent of the 20 assay stories, this screen does not select voxels from an evaluation outcome. The resulting estimand remains held-out-story prediction within UTS03; the retained voxels are spatial measurements, not independent biological replicates.

Table 6: Contextual and nuisance feature blocks by assay.

Assay	Contextual block	Low-level nuisance block	Static or additional block
Tuckute	Mean-pooled activations of the declared LM layer over non-padding token positions	Sentence token length and item position, the zero-based index divided by one less than the sentence count	Mean noncontextual input embedding. Released imageability ratings enter the Tuckute assays only as a labeled sensitivity, except the target-projection assay, where they are primary
LeBel	Final-subtoken hidden state for each word under its available left context	Word rate, phoneme rate, word duration, word length, and log word frequency	the released `english1000` lexical-semantic basis (985 dimensions)

B.2 Stimulus, nuisance, and fold-local preprocessing

Each assay separates the contextual model feature from lower-level or static stimulus features that could explain recorded responses without the contextual representation under study. Table 6 summarizes these blocks before the fold-specific transformations are applied.

In the controlled regional assay, the original implementation recomputed the static embedding block for each trained or untrained LM. The trained score remains controlled relative to its implemented nuisance set, but the trained-minus-untrained gap does not isolate learned parameters against a byte-identical static nuisance block. The saved-student biological-transfer assay instead fixes the static block to GPT-2 Medium so that every student arm is evaluated against the same nuisance representation. The 50-component linear target-projection assay adds the released imageability covariate in its primary design; Appendix F reports the design without imageability as a continuity sensitivity. GPT-2 XL surprisal and probabilistic context-free-grammar surprisal provide a secondary stress test because both contain model-derived contextual information that overlaps the construct being measured.

The Tuckute analyses use five contiguous outer folds of 200 sentences. For each fold, stimulus-domain scaling and contextual and static PCA are fitted on the other 800 sentences and applied unchanged to the held-out block. Contextual and static feature blocks are each reduced to rank 50, equalizing their nominal dimensions before readout fitting, while the scalar nuisances remain unprojected. The synthetic brain-response target has one fixed exception. Its coordinate standardization is learned once from the external WikiText training corpus, as described below. Any target PCA used in a Tuckute assay is still fitted only on the corresponding 800 outer-training rows.

The LeBel preprocessing begins with the time-aligned word and phone annotations in the TextGrid files. Word duration is the successive word-onset difference clipped to 0–3 s, with 0.3 s assigned to the terminal word. Contextual word features are extracted in overlapping 256-token windows with stride 128; each word receives the hidden state of its final subtoken under the maximum available left context. Word length, log word frequency, word duration, and `english1000` features are Lanczos-resampled with window 3 to the 2.0045-s grid. Word and phoneme rates are direct counts in TR-centered histogram windows.

After these stream-specific operations, the first 10 and final 5 samples of each story are removed with the fixed slice `[10:-5]`. The contextual, low-level, and static feature streams then receive within-story z-scoring and finite-impulse-response copies at delays one through four TRs. The recorded blood-oxygen-level-dependent (BOLD) responses are z-scored voxelwise within story but receive no finite-impulse-response expansion. For a held-out story, this transformation uses that story's own response mean and variance before scoring. This is a shared transductive normalization across model conditions, so the estimand is prediction of within-story standardized responses rather than deployment to a story whose response moments are unavailable. Feature and response rows are then truncated to their common aligned length. Within each outer fold, the contextual and static blocks receive train-story PCA to rank 100, while the low-dimensional nuisances remain unprojected. A predictor scaler fitted on the training stories then z-scores the assembled design. The same folds, nuisance arrays, temporal transforms, and response rows are used for trained and architecture-matched untrained LMs. Appendix D specifies response centering, ridge selection, fresh-readout fitting, aggregation, and statistical inference.

B.3 Synthetic brain responses and text-derived auxiliary targets

Target construction. The synthetic-response route uses a fixed WikiText-103 corpus that is disjoint from the Tuckute sentences. The corpus builder reads the training split of `Salesforce/wikitext`, configuration `wikitext-103-raw-v1`, in dataset order. It removes blank lines and document headers, splits each remaining line after sentence-final punctuation, retains sentences of 4–45 whitespace-delimited words, and removes case-insensitive duplicates and exact case-insensitive matches to Tuckute sentences. The first retained sentences form the fixed training file and the following sentences form the untouched held-out file. The build retains 96,000 training and 2,000 held-out sentences, which the distillation-headroom analysis uses directly. The synthetic-target route reads the same two files but skips one blank line in each, so its caches hold 95,999 and 1,999 sentences; both pairs of counts describe the same split.

TRIBE v2 maps each sentence to synthetic brain responses on a standardized cortical surface with 20,484 fsaverage5 locations [27]. The target builder loads `facebook/tribev2`, sets `data.text_feature.model_name` to `unsloth/Llama-3.2-3B`, and sets `data.features_to_use` to `[text]` in the runtime configuration. The Llama-3.2-3B component is TRIBE's internal text encoder; it is distinct from the frozen GPT-2 Medium distillation teacher. The published TRIBE description does not list the Tuckute sentence set among its training datasets. This supports dataset separation at the level of reported provenance, but it is not a categorical proof that no overlapping text or derivative data entered any upstream component.[27] For this thesis, the sentence-level target is the average of TRIBE's one-second output samples for that sentence. This operation creates one training vector per sentence; it is not a claim of physical alignment to the acquisition timing of the Tuckute fMRI responses. Its `synthetic-text` configuration assigns each word a 0.35-s event followed by a 0.05-s gap, lengthens the final event of any timeline shorter than 1.0 s until the timeline reaches 1.0 s, and uses a 1.0-s TR. Each sentence has a separate timeline, and its retained one-second predictions are averaged coordinatewise to form one 20,484-dimensional vector. The same construction is applied to Tuckute item IDs 1–1,000, with `item_id=item_index+1`.

Let Y_{ij} be synthetic response coordinate j for training sentence i. Each coordinate is standardized using the training-corpus mean $\mu_{j,\text{train}}$ and standard deviation $\sigma_{j,\text{train}}$.

$$\widetilde{Y}_{ij} = \frac{Y_{ij} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}} + 10^{-6}}.$$

This operation centers and scales each cortical coordinate using only the synthetic-response training corpus. The same stored moments are applied to held-out sentences, and the small constant prevents division by a near-zero scale. The moments and standardized targets are computed and stored in float32. The stored training target retains all 20,484 standardized coordinates. Appendix C specifies how the training objective uses them; no target PCA is applied before that use. When a later assay requires a lower-dimensional target, its PCA basis is learned only from the relevant outer-training rows.

The text-derived auxiliary target preserves the target width and standardization procedure while replacing the TRIBE construction with frozen LM features. Let $h_i \in \mathbb{R}^{1024}$ be the mean-pooled layer-12 representation of sentence i extracted from the same frozen GPT-2 Medium model used as the distillation teacher, with at most 64 token positions. Let $R \in \mathbb{R}^{1024\times 20484}$ be drawn once with `numpy.random.default_rng(0)`, with entries $R_{ab} \sim \mathcal{N}(0,1)$, and then frozen. The target for sentence i is

$$T_i = \frac{h_i R}{\sqrt{1024}}.$$

The projection gives the auxiliary target the same 20,484-coordinate width as the synthetic brain-response target. The factor $1/\sqrt{1024}$ keeps its scale from growing mechanically with the 1,024 input dimensions. Each projected column is standardized by the same formula, using its own mean and standard deviation computed on the WikiText training corpus.

Pairing controls. The controls differ in which relationship they disrupt. The following constructions make the preserved and disrupted relationships explicit.

The recorded-response intervention constructs its permutation only from the current 800-sentence training complement. The participant-averaged route uses one draw per fold and seed, whereas the participant-specific route fits five independently permuted controls per fold and seed and averages their scores; that replicate count is a fixed choice and is unrelated to the five ROIs. The control

Table 7: Synthetic targets and pairing controls used in the thesis.

Construction	Matched or preserved property	Attribution role
Synthetic brain-response target	Sentence order, TRIBE geometry, standardization, and width	Synthetic-response supervision under study
Text-derived auxiliary target	Sentence order, width, standardization, and corpus	Frozen GPT-2 Medium projection controlling for auxiliary supervision derived from text
Recorded-response permutation	Response values, marginal distribution, and within-block order	Randomly reordered sentence–response pairing; fixed blocks remain possible
Saved block-permutation sensitivity	Synthetic target values and approximate block structure	Earlier pairing sensitivity only; fixed, duplicated, or omitted rows remain possible
Frozen row-twin control	Exact within-block row multiset and train-only 50-component PCA capacity	Row identity under one deterministic derangement with no fixed points

randomly reorders ten contiguous response blocks within the 800-sentence training complement. It disrupts sentence–response pairing while preserving the response-value multiset and within-block order, but it is not constrained to move every block.

The saved synthetic block permutation and the frozen row twin serve different purposes. The earlier saved block permutation split the rows into ten contiguous blocks, permuted the block order, and copied each source block into its destination block only up to the shorter of the two lengths. Because the blocks are unequal, a destination row past that cut keeps its own target and a source row past that cut is never used, so the result can retain correctly paired rows and can duplicate or omit a row. It is therefore only a sensitivity. The saved run does not record an exact fixed-row count, so the thesis neither quantifies that count nor treats this control as an identifying comparator. A derangement is a permutation with no fixed points, so every stimulus is paired with another stimulus's target. The frozen row-twin control instead uses Sattolo's algorithm to construct one derangement inside each 200-item outer block before target values or participant scores are read. Its deterministic seed is derived from the block identifier. For each outer fold, aligned TRIBE rows define one train-only 50-component PCA basis, and the same basis is applied to the twin. This comparison isolates row identity at matched target-row content and linear target capacity. It concerns one frozen row-twin control and therefore does not establish invariance across possible derangements or identify a causal brain mechanism.

Table 7 summarizes the construction and attribution role of each target or control.

The text-derived auxiliary target matches the synthetic target on row order, output width, standardization procedure, and corpus. The two targets do not share a source representation, and matching width and standardization does not establish matched covariance, effective rank, recoverability, or optimization difficulty. Those differences limit the attribution question the comparison can answer; Section 4.5 and Appendix F report the measured audit. With the data, targets, and controls fixed, Appendix C specifies how the training objectives use them.

C Training Interventions and Reproduction

This appendix specifies the optimized objectives, the parameters reached by each gradient, and the student state available after training. Appendix B defines the data, targets, and controls; Appendix D defines the fresh readouts and statistical inference; and Section 4 with Appendices E–F report the resulting observations and verdicts. Temporary target heads never enter a later evaluation.

C.1 Ordinary KD and language-quality rules

Let $z_T(x)_t$ and $z_S(x)_t$ be teacher and student logits at a non-padding causal-language-model position t. At temperature τ, their softened token distributions are

$$p_T^\tau(\cdot \mid x, t) = \mathrm{softmax}(z_T(x)_t/\tau)\,, \qquad q_S^\tau(\cdot \mid x, t) = \mathrm{softmax}(z_S(x)_t/\tau)\,.$$

Table 8: Language objectives and quality rules by training family.

Family	Teacher, student, and initialization	Language objective and corpus	Quality rule
Distillation-headroom analysis	GPT-2 Medium teacher; GPT-2 Small student initialized either from pretrained weights or randomly	Output KD on 96,000 WikiText sentences; 2,000 fixed held-out sentences	Held-out perplexity is a convergence measure; the warm and random-start routes differ in initialization and training duration
Recorded-response intervention (sentence)	Qwen2.5-0.5B student; Qwen2.5-1.5B teacher when the route is teacher-anchored	Masked next-token cross-entropy in the unanchored screen; output KD in anchored routes; fixed WikiText quality probe	The recorded-response arm and permuted-response control share the scoring implementation; ordinary KD comparisons require perplexity within the tolerance stated in Section 3.5
Recorded-response intervention (naturalistic)	Qwen2.5-0.5B student; frozen Qwen2.5-1.5B teacher in anchored routes	Unanchored or output-KD-anchored language path, according to the executed grid	The full-parameter route required improved held-out predictivity without a collapse in language quality on the fixed probe; no numeric maximum was fixed in advance
Synthetic-response intervention	GPT-2 Medium teacher; pretrained GPT-2 Small student shared within seed	Output KD on 95,999 WikiText sentences; 1,999 fixed held-out sentences	Each target arm must satisfy Equation 15; saved-student comparisons also apply the within-seed bits-per-byte tolerance of Section 3.5

The implemented output-distillation loss is

$$\mathcal{L}_{\mathrm{KD}} = \tau^2 \frac{\sum_{x,t} m_{xt} D_{\mathrm{KL}}(p_T^{\tau}(\cdot \mid x,t) \,\|\, q_S^{\tau}(\cdot \mid x,t))}{\sum_{x,t} m_{xt}}, \tag{24}$$

where m_{xt} is one at non-padding positions and zero otherwise. Equation 24 averages teacher-to-student token-level KL divergence over valid positions. The factor τ^2 compensates for the gradient-scale change introduced by temperature softening. Every executed output-distillation route uses $\tau = 2$, gradient-norm clipping at 1, and AdamW at its library default decay rates and weight decay, with a constant learning rate and no warmup. Two route types recur below. An unanchored route trains the student without a teacher and keeps language quality with its own next-token objective, and a teacher-anchored route adds output KD from a frozen teacher.

The language objective is shared within a paired intervention, but the student initialization and the quality rule that applies differ across families. Table 8 records those differences.

The headroom family uses seeds 0–2, batch size 16, and maximum length 64. The pretrained student is trained for one epoch at 5×10^{-5}, whereas the random-start student is trained for two epochs at 3×10^{-4}. These are separate optimization regimes, not an initialization-only contrast. The recorded- and synthetic-response settings are specified with their target paths below.

C.2 Recorded-response intervention

For a recorded response y_i, the common sentence-level target path is

$$x_i \longrightarrow h_{12}(x_i) \longrightarrow W h_{12}(x_i) \longrightarrow \mathcal{L}_{\mathrm{target}}.$$

Here h_j denotes entry j in the model's returned hidden-state sequence, whose entry 0 is the embedding output. The thesis shorthand "hidden-state index 12" therefore names the exact attachment returned by the implementation. The vector $h_{12}(x_i)$ is mean-pooled over non-padding token positions, and W is a temporary linear map from the pooled hidden state to the five-ROI response vector of Appendix B, unless the selected loss is readout-free.

The common joint objective is

$$\mathcal{L} = \lambda_{\text{lang}}\mathcal{L}_{\text{lang}} + \lambda_{\text{target}}\mathcal{L}_{\text{target}}, \qquad \lambda_{\text{lang}} = 1. \tag{25}$$

The first term preserves the declared language objective, while the second exposes the retained student path to the recorded response. For teacher-anchored routes, $\mathcal{L}_{\text{lang}} = \mathcal{L}_{\text{KD}}$; the unanchored screen instead uses masked next-token cross-entropy.

For predictions $\widehat{Y}$ and targets Y with minibatch size B and target width D, the squared-error target loss is

$$\text{MSE}(\widehat{Y}, Y) = \frac{1}{BD}\sum_{b=1}^{B}\sum_{j=1}^{D}\left(\widehat{Y}_{bj} - Y_{bj}\right)^2.$$

This loss averages equally over minibatch rows and target coordinates. Consequently, every reported target weight is interpreted under this normalization.

The common sentence-level regime places rank-16 LoRA adapters with scaling 32 and dropout 0.05 in the attention and feed-forward projections of the student family, namely `c_attn`, `c_proj`, and `c_fc` for GPT-2, and `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` for Qwen2.5 [29]. The pretrained base, input embeddings, and tied output matrix are frozen; the adapters and a co-trained W are optimizer parameters. The target gradient reaches W and the adapters that produce h_{12}, while adapters after the attachment state receive only the language-loss gradient. After training, the adapters are merged into the student and the temporary target map is discarded. The merged student is frozen before a fresh evaluation readout is fitted as specified in Appendix D.

Table 9 states only the implementation change relative to this common path.

The unanchored sentence screen also explored alternative target losses, holding the co-trained MSE grid fixed and changing only $\mathcal{L}_{\text{target}}$. Cosine and squared-Pearson losses use a co-trained target map. Linear CKA is readout-free. It compares the minibatch geometry of h_{12} and the response matrix directly, so no temporary W participates in that branch [30]. These are implementation variants of the recorded-response intervention; their outcomes remain in Section 4.3 and Appendix E.

The practical-utility evaluation used this same recorded-response training path, with a Qwen2.5-0.5B student against a Qwen2.5-1.5B teacher at h_{12}, seeds 0–7, target weight 10, 800 training items, three epochs, learning rate 2×10^{-4}, and batch size 16, then 400 items scored from each of the two out-of-domain corpora of Section 3.5.

C.3 Synthetic-response intervention

The synthetic-response intervention uses the same joint objective in Equation 25, but the attachment and trainable path differ. For GPT-2 input x, the retained student path is

$$x \longrightarrow e_{\text{tok}}(x) + e_{\text{pos}}(x) \longrightarrow h_6(x) \longrightarrow h_{7\text{–}12}(x) \longrightarrow q_S^{\tau}(\cdot \mid x),$$

and the temporary target branch is

$$\text{meanpool}(h_6(x)) \longrightarrow W_{\text{tmp}}\,\text{meanpool}(h_6(x)) \longrightarrow \mathcal{L}_{\text{target}}.$$

As above, h_6 names hidden-state index 6 in the returned sequence, not an informal count of visible blocks. The output-KD gradient reaches the entire student path. The target gradient reaches W_{tmp}, token and positional embeddings, normalization components, and student blocks that produce h_6; blocks after the attachment receive only the output-KD gradient. Because GPT-2 ties its token embedding and output matrix, that retained matrix receives the target gradient through its input role and the output-KD gradient through both roles.

The teacher is frozen, whereas every student parameter and W_{tmp} is optimized. All target arms map the 768-dimensional mean-pooled attachment state to all 20,484 coordinates using the same mean MSE. Appendix B defines the construction and standardization of the synthetic brain-response target, the text-derived auxiliary target, and the saved block permutation. The common optimization grid uses 95,999 training sentences, one epoch, learning rate 5×10^{-5}, batch size 4, and maximum length 64.

Table 9: Recorded-response intervention variants.

Variant	Change from the common target path	Executed grid	Retained object
Co-trained mean squared error (MSE)	A new W is trained jointly with the producing adapters; correctly paired and permuted responses use the same path	Rank 16; seeds 0–2; 3 epochs; 2×10^{-4}; batch 16; target weight 10. Participant-averaged training uses one permutation draw; participant-specific training uses five	Merged student; W discarded
Frozen readout	W_0 is ridge-fitted on untuned outer-training features with $\alpha = 100$ and remains fixed during intervention training	Same grid and paired-permutation construction as the co-trained MSE row	Merged student; W_0 discarded
Contrastive	Symmetric InfoNCE compares normalized Wh_{12} with the paired five-ROI response using in-batch negatives [31]	Rank 16; seeds 0–1; 3 epochs; target weight 1; temperature 0.07; three permutation draws	Merged student; W discarded
Increased capacity	The co-trained MSE path is unchanged; adapter rank increases from 16 to 64	Seeds 0–1; 6 epochs; target weight 10; five permutation draws; otherwise as in the participant-specific grid	Merged student; W discarded
Naturalistic voxelwise	A participant-specific head maps pooled 25-word segments to reliable voxels after a 4-s response delay	UTS01–03; 14 tuning and 6 evaluation stories. The unanchored grid ran all three participants at seeds 0–1, two permutation draws, and target weight 10. The output-KD-anchored grid swept target weights 1, 3, and 6 on UTS03 over the 2,801 voxels whose repeat reliability exceeds 0.7	Merged student; voxel head discarded
Full parameter	The voxelwise MSE and output-KD anchor update all student parameters except the tied input/output embedding; no LoRA is used	UTS01–03; seeds 0–2; 2 epochs; 10^{-5}; batch 16; target weight 10	Student used for immediate frozen evaluation; voxel head discarded; no reusable student checkpoint retained

Table 10: Synthetic-response training arms.

Arm	Additional supervision	Shared optimization settings
Ordinary KD arm	None; $\lambda_{\text{target}} = 0$ and no temporary target map	Pretrained GPT-2 Small initialization, frozen GPT-2 Medium teacher, corpus, output-KD loss, and seeds 0–5
Synthetic-response arm	Correctly paired synthetic brain-response target through W_{tmp}; $\lambda_{\text{target}} = 0.1$	Same initialization, teacher, corpus, schedule, and seeds 0–5 as the matched ordinary KD arm
Saved block-permutation sensitivity	Saved block-permuted synthetic target and saved block-permuted text-derived target, each through the same W_{tmp} and mean MSE	Same initialization, teacher, corpus, schedule, target weight, and seeds 0–5
Text-derived auxiliary-control arm	Text-derived auxiliary target through the same target-map shape and mean MSE; $\lambda_{\text{target}} = 0.1$	Same pretrained student, teacher, corpus, and schedule; executed for seeds 0–5

Table 11: Training entry points and retained replay information.

Family	Executed entry point	Final-state and retained-output rule
Distillation-headroom analysis	`scripts/run_kd_alignment.py`	Final warm or random-start student evaluated in memory; the run records its configuration and quality measurements
Recorded-response intervention (sentence)	`scripts/run_brain_lever.py`; target losses in `scripts/brain_loss.py`	Final merged student evaluated after each fold–seed cell; temporary target map discarded; no intermediate quality frontier retained
Recorded-response intervention (naturalistic)	`scripts/run_lebel_tune.py`	Final merged or full-parameter student evaluated immediately; temporary voxel head discarded
Synthetic-response intervention	`scripts/run_tribe_phase3.py`	Final student evaluated in memory; optional checkpoint written only when a save directory is supplied; temporary target map discarded

The text-derived auxiliary-control arm was first run for seeds 0–2 and extended to seeds 3–5 before any participant outcome was scored; all reported analyses use the full six seeds.

At the final optimizer step, the temporary target map is always discarded. Whether a run evaluates its student immediately or a later assay reloads a saved checkpoint, every post-training score comes from a readout fitted afresh on the frozen student. Training can therefore shape the retained student through the target gradient, but the training head itself never computes a target-recovery or biological-transfer score.

The frozen row-twin control is not a training arm. It is constructed only for the later 50-component linear target-projection assay, and Appendix B specifies how it is constructed. Comparator adequacy and all arm outcomes are reported in Section 4.5 and Appendix F.

C.4 Reproduction scope

Reproducing a run requires four linked objects: the locked software environment, the executed entry point and arguments, the declared seed and final-state rule, and the retained output. Table 11 indexes the first three; each experiment's own record states the exact configuration.

The environment is defined by the Python requirement in `pyproject.toml` and the exact package and CUDA build resolved by `uv.lock`. These pin the current environment rather than reconstruct the one every earlier run used. Each run stores the model identifiers and training configuration it executed. Many early runs, however, do not record an immutable upstream Hugging Face revision, so exact replay of those runs additionally depends on a retained model cache or a saved student. This is a real limit on reproduction and is reported rather than filled with an inferred revision.

With the optimization and reproduction scope fixed, Appendix D specifies how the retained students are evaluated and how uncertainty is assigned.

D Estimators, Aggregation, and Inferential Scope

Section 3.5 states the design-level estimands and inference units. This appendix answers four narrower questions: how the predictive quantities are related, how fitting and aggregation are nested, how uncertainty and decision rules are assigned, and where information-theoretic analogies stop. Appendix B defines the inputs and preprocessing, Appendix C defines the training objectives and retained model states, and Section 4 reports the empirical outcomes. The quantities below are assay-conditional predictive contrasts. They are neither causal effects nor direct estimates of information-theoretic objects.

D.1 Predictive estimands and ridge estimation

For response coordinate j, Section 3.2 defines the held-out score $R_j^2(M)$ for a readout design M in Equation 1. Let N denote the declared nuisance design and $F = [N, X]$ the full design after adding contextual LM representations X. The nuisance-only and full scores are $R_j^2(N)$ and $R_j^2(F)$, and the reported semipartial increment is

$$\Delta R^2_{\mathrm{semi},j}(X) = R_j^2(F) - R_j^2(N).$$

This difference asks how much held-out predictive value is gained by adding X to the nuisance design. Because the nuisance-only and full ridge readouts are separately regularized and evaluated out of sample, the difference can be negative. It is not a nonnegative variance decomposition.

Population partial R^2 answers a different question. Let $\mathcal{R}^2_{N,j}$ and $\mathcal{R}^2_{F,j}$ be the population coefficients of determination for nested, unregularized linear projections on N and F. When $\mathcal{R}^2_{N,j} < 1$,

$$R^2_{\mathrm{partial},j} = \frac{\mathcal{R}^2_{F,j} - \mathcal{R}^2_{N,j}}{1 - \mathcal{R}^2_{N,j}} = 1 - \frac{\mathrm{Var}(Y_j - \Pi_F Y_j)}{\mathrm{Var}(Y_j - \Pi_N Y_j)}, \tag{26}$$

where Π_N and Π_F are the corresponding population linear projections. Equation 26 measures the fraction of variance left after the nuisance projection that is removed by adding X. The semipartial quantity instead measures the gain relative to total outcome variance. The thesis reports the cross-validated semipartial increment, not a transformed cross-validated partial coefficient.

Aggregation across target or response coordinates defines another distinction. For J held-out coordinates, let SSE_j and SST_j denote the coordinate-specific residual and total sums of squares. The two aggregations used in the thesis are

$$\overline{R^2}_{\mathrm{equal}} = \frac{1}{J}\sum_{j=1}^{J}\left(1 - \frac{\mathrm{SSE}_j}{\mathrm{SST}_j}\right), \qquad R^2_{\mathrm{pooled}} = 1 - \frac{\sum_{j=1}^{J}\mathrm{SSE}_j}{\sum_{j=1}^{J}\mathrm{SST}_j}. \tag{27}$$

The first expression gives every coordinate equal weight. The second weights coordinates by their held-out variance, so a nearly constant coordinate cannot contribute as much as a variable coordinate. The WikiText synthetic-target recovery assay uses equal-coordinate averaging, whereas the saved-student target-retention analysis pools sums of squares across the fixed target coordinates. Neither calculation treats coordinates as independent repetitions.

Ridge regression supplies a regularized linear predictor. Let $y \in \mathbb{R}^n$ be a centered response vector, let $M \in \mathbb{R}^{n\times p}$ be the centered and standardized predictor matrix, and let $\beta \in \mathbb{R}^p$ be its coefficient vector. If

$$y = M\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \qquad \beta \sim \mathcal{N}(0, \tau_\beta^2 I),$$

then the posterior mode minimizes

$$\|y - M\beta\|_2^2 + \lambda\|\beta\|_2^2, \qquad \lambda = \frac{\sigma^2}{\tau_\beta^2}.$$

This is a maximum a posteriori interpretation only under the stated Gaussian likelihood, isotropic zero-mean prior, and scaling convention. The fitted linear predictor represents the conditional mean only when that mean is linear and the model is correctly specified.

The sentence-level Tuckute and WikiText readouts use the candidate penalties

$$\{1, 10, 100, 1000, 10000\}.$$

Each multi-output fit selects one scalar penalty on its training partition by generalized cross-validation, the closed-form leave-one-out criterion, rather than by an explicit fold split; it does not select a separate penalty for every response or target coordinate. The naturalistic LeBel readouts use

$$\{10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7\},$$

again selecting one multi-output penalty from training stories only.

The comparator-adequacy audit uses

$$\{0.1, 1, 10, 100, 1000\}$$

and selects one penalty by aggregate held-out R^2 across both target families and both fixed coordinate samples within five contiguous inner folds. These selection rules are nested inside the corresponding training partition; the untouched outer block or held-out corpus is used once for evaluation. In the executed assays, λ is selected from a finite grid using the relevant training complement; operationally, it is predictive regularization rather than a fixed scientific prior. Neither this derivation nor a positive held-out score makes the fitted coefficients causal, identifies a biological mechanism, or shows that an LM and the brain implement the same computation.

D.2 Cross-validation, aggregation, and inference units

Under the governing leakage rule, every data-adaptive operation that can use stimulus-domain information is fitted on the relevant training complement. Fixed protocol operations may be applied before the split, but learned centering, scaling, dimensionality reduction, penalty selection, and readout fitting remain inside it. Appendix B specifies the exact fixed preprocessing and split definitions.

The naturalistic assays require one explicit distinction. Feature resampling, edge slicing, within-story feature and BOLD z-scoring, and finite-impulse-response delay copies are fixed story-local operations. Because the held-out story's response normalization uses that story's own moments, the resulting estimand is conditional on this normalization protocol. Across-story PCA, assembled-design scaling, response centering, ridge selection, and readout fitting are learned only from the readout-training stories. The 50-component linear target-projection assay has one other declared exception. It reuses target-standardization moments fixed during target construction, while its target PCA and readouts remain fold-local.

Table 12 records the held-out boundary, learned operations, and final inference unit for each scientific role.

The aggregation order can be written without treating technical repetitions as new scientific units. Let $\mathcal{C}_u$ be the set of technical cells contributing to final unit u, such as its seeds, folds, or seed–fold combinations, let $d_{u,c}$ be the matched contrast in cell c, and let $\mathcal{A}_u$ denote the declared within-unit aggregation.

$$\widehat{\theta}_u = \mathcal{A}_u(\{d_{u,c}\}_{c\in\mathcal{C}_u}), \qquad \widehat{\Theta} = \mathcal{I}\Big(\widehat{\theta}_1, \ldots, \widehat{\theta}_U\Big). \tag{28}$$

Here $\mathcal{A}_u$ may be a mean, a median, or pooled sums of squares, while $\mathcal{I}$ is the descriptive summary or inferential procedure applied to the U final units. Equation 28 says that seeds and folds are first reduced within a participant when the participant is the scientific unit. Analyzing every seed–fold cell as if it were an independent participant would change both the estimator and its uncertainty.

The final unit determines what can generalize. Seeds vary optimizer randomness under one fixed procedure. Stimulus folds share observations and have overlapping training complements. Voxels and ROIs share a participant, preprocessing, and stimuli. Target coordinates share one generated target and student. None can replace a participant when the claim concerns people [23]. Conversely, several participants evaluated on one fixed sentence set do not by themselves establish generalization to a population of stimuli. Shared-fold and story-block summaries are therefore robustness checks, not substitutes for participant or stimulus-population inference.

D.3 Uncertainty, multiplicity, and decision rules

Four choices must remain separate. The inference unit identifies the observations that enter uncertainty. The interval or test describes variation across those units. The multiplicity family identifies which simultaneous claims require adjustment. The decision rule states what evidence was required to continue to a later stage. Unless stated otherwise, the reported intervals are two-sided 95% intervals.

For U final-unit estimates $\widehat{\theta}_1, \ldots, \widehat{\theta}_U$, the ordinary t interval is

$$\overline{\theta} \pm t_{1-\alpha/2,U-1}\frac{s_\theta}{\sqrt{U}}, \qquad \overline{\theta} = \frac{1}{U}\sum_{u=1}^{U}\widehat{\theta}_u. \tag{29}$$

Here α is the two-sided error rate and s_θ is the sample standard deviation of the U final-unit estimates. Equation 29 uses the number of final units U, not the number of coordinates, folds, or technical runs. Where a table reports a descriptive interval over technical seeds, it applies the same formula to the

Table 12: Cross-validation nesting and inference units by scientific role.

Scientific role	Held-out boundary	Operations learned inside the boundary	Aggregation and final unit
Controlled regional predictivity assay	Five contiguous Tuckute folds, each with an 800-sentence training complement and a 200-sentence test block	Centering, scaling, PCA, one multi-output ridge penalty, and the response readout	Declared ROIs and folds are averaged; the sentence set is fixed, and untrained-model seeds are technical repetitions
Distillation-headroom analysis	The same five Tuckute folds for brain predictivity and a fixed WikiText probe for language quality	Brain-response transforms and readouts are fitted inside each fold; quality scores use the fixed scoring protocol	Model condition is the descriptive unit, seeds index stochastic arms, and model families are the resampling clusters for the cross-family quality association
Recorded-response intervention	The student trains on each 800-sentence outer complement; the untouched 200-sentence block is divided into five inner readout folds	Inner-fold transforms, penalty selection, and the fresh response readout	Participant-averaged analyses end at five shared outer folds; participant-specific analyses aggregate seed–fold cells within each participant
Controlled naturalistic voxelwise predictivity assay	Five complete-story folds; voxel reliability uses a separate repeated story	Training-story PCA, assembled-design scaling, response centering, one multi-output ridge penalty, and the readout	Story-block signs and voxel summaries describe within-participant robustness; they do not create participant inference
Naturalistic recorded-response intervention	Complete stories remain held out from fresh evaluation readouts	Training-story transforms, ridge selection, and participant-specific voxel readouts	Seeds are technical repetitions within three descriptive participant probes
Synthetic-target recovery assay	The student trains on 95,999 WikiText sentences and is evaluated on 1,999 fixed held-out sentences	The fresh target-readout penalty is selected using training data	Equal-coordinate target R^2 is compared within six matched technical seeds
Saved-student biological-transfer assay	Frozen students enter five contiguous Tuckute 800/200 response-readout folds	All representation transforms, nuisance transforms, penalties, and fresh response readouts use the 800-sentence complement	Six seeds and five folds are aggregated within each of nine participants; participants are the biological units
50-component linear target-projection assay	Five contiguous Tuckute folds; aligned targets and the frozen row twin share the train-fold target-PCA basis	Each aligned or twin multi-output readout selects one scalar penalty on its own training complement	Folds are aggregated within each of nine participants; there is no student-seed axis
Saved-student target-retention analysis	Frozen students and targets use the same five outer folds	Representation and target transforms and the multi-output target readout remain train-fold-local	Sums of squares are pooled across target coordinates; paired contrasts end at six technical seeds
Composed-path diagnostic	Both linear paths are fitted on the same outer-training rows and scored once on the untouched test block	Target-to-response and student-to-target coefficient blocks are fitted separately inside each fold	Folds and six seeds are aggregated within each of nine participants
Practical-utility evaluation	Fixed in-domain and out-of-domain corpora are scored without fitting a biological readout	No evaluation-time model fitting; corpus-level losses are computed under the fixed scoring protocol	Paired arm contrasts end at eight technical seeds and remain conditional on the fixed models and corpora

K matched seed contrasts below and is read as a stability summary of one frozen procedure, not as a sampling interval over scientific units. Its sampling interpretation requires independent final-unit estimates and approximate normality when U is small. Participant intervals are conditional on the fixed stimuli, trained students, and analysis protocol used to construct each participant estimate.

For K matched technical contrasts $d_1, \ldots, d_K$, the exact two-sided magnitude-preserving sign-flip test used in the seed-level analyses is

$$p_{\text{flip}} = \frac{1}{2^K} \sum_{\epsilon \in \{-1,+1\}^K} \mathbf{1}\left[\left| \frac{1}{K} \sum_{k=1}^{K} \epsilon_k d_k \right| \geq \left| \frac{1}{K} \sum_{k=1}^{K} d_k \right| \right]. \tag{30}$$

Under the null hypothesis that each matched contrast is symmetrically distributed about zero, the contrast signs are exchangeable conditional on their magnitudes, which is what this test requires. It differs from the participant sign test, which uses only the number of positive and negative participant effects and requires independent participant signs. The Wilcoxon signed-rank sensitivity additionally

Table 13: Uncertainty procedures and decision rules by scientific role.

Scientific role	Interval or test and its unit	Multiplicity and decision scope
Controlled regional predictivity assay	Held-out-fold dispersion and trained–untrained contrasts on the fixed sentence set	Prespecified layer and nuisance grids bound the assay; folds do not support participant or stimulus-population inference
Controlled naturalistic voxelwise predictivity assay	Within-participant trained–untrained point contrasts and five story-block signs; no voxel-resampling interval	Spatial voxels are not independent units, and overlapping story folds remain robustness checks
Distillation-headroom analysis	Fold–seed bootstrap conditional on convergence; cross-family quality association uses a family-cluster bootstrap, with pooled Fisher intervals only as a sensitivity	The declared retention bands classify opportunity within this assay and are not universal alignment thresholds
Participant-averaged recorded-response intervention	Seeds are averaged within each outer fold before a five-fold t interval	The interval describes stability across the five shared blocks of the fixed sentence set
Participant-specific recorded-response intervention	Participant medians over seed–fold cells receive a participant t interval and exact one-sided sign test; a shared-fold interval and cluster bootstrap assess block sensitivity	Requiring agreement between participant and shared-fold readings is a conservative conjunctive rule, not a joint crossed-population confidence interval
Naturalistic recorded-response intervention	Seed summaries are technical repetitions within three participant-specific probes	The small participant set supports descriptive mechanism probing, not population inference
Synthetic-target recovery assay and retained-student movement audit	Six matched-seed contrasts receive descriptive intervals and declared exact sign-flip tests	Manipulation and language-quality criteria qualify later interpretation but provide no biological replication
Comparator-adequacy audit and attribution assessment	Fixed-sample margins and paired arm contrasts are reported without coordinate-level p-values	Comparator adequacy is a pre-outcome or baseline design condition for attribution; realized movement is diagnostic, and statistical precision cannot repair a non-identifying comparator
Saved-student biological-transfer assay	Nine participant effects receive a t interval, a one-sided upper bound, an exact sign test, and a Wilcoxon sensitivity	Seeds and blocks remain descriptive; the upper bound belongs to the relative contrast against the text-derived arm and is compared only with the continuation rule fixed before the results were seen
50-component linear target-projection assay	Participant effects receive t intervals and exact sign tests	Its participant quantities belong to a separate Bonferroni family with a simultaneous one-sided bound
Saved-student target-retention analysis	Six paired technical-seed contrasts receive a descriptive interval and an exact 2^6 sign-flip test	Target retention has its own adjusted interval family and does not create biological evidence
Composed-path diagnostic	Participant path-product quantities receive t intervals and exact sign tests	The composed path has its own adjusted interval family; it is predictive and is not a causal mediation effect
Practical-utility evaluation	Eight paired technical-seed endpoint observations were recorded, but the completed analysis reports only mean contrasts and an observed-variance sensitivity calculation	No paired endpoint interval or equivalence test was retained; the MDE supplies no hypothesis or equivalence decision

assumes independent and approximately symmetric paired effects. The shared-fold cluster bootstrap resamples the five outer-fold summaries as clusters. Because the folds reuse observations through overlapping training complements, both that bootstrap and the fold t interval are fixed-block stability summaries rather than stimulus-population intervals. The cross-family quality bootstrap instead resamples model families rather than individual models.

Table 13 maps each scientific role to its uncertainty procedure and stage-specific decision scope.

For the primary retained-student movement audit, let H_s be the layer-6 representation matrix of the retained synthetic-target student and H_s^{KD} that of its seed-matched ordinary-KD student, both on the same $n = 1{,}999$ WikiText sentences held out from every training arm in technical seed s, and let $C = I - \mathbf{1}\mathbf{1}^\top/n$ center examples. The centered relative-Frobenius distance and linear-CKA distance are

$$D_{F,s} = \frac{\|C(H_s - H_s^{\mathrm{KD}})\|_F}{\|CH_s^{\mathrm{KD}}\|_F}, \qquad D_{\mathrm{CKA},s} = 1 - \frac{\|(CH_s)^\top (CH_s^{\mathrm{KD}})\|_F^2}{\|(CH_s)^\top (CH_s)\|_F \, \|(CH_s^{\mathrm{KD}})^\top (CH_s^{\mathrm{KD}})\|_F}. \tag{31}$$

The first quantity measures centered activation change relative to the ordinary-KD activation norm. The second measures the change in centered linear representation geometry, with zero denoting identical linear-CKA geometry. Global parameter displacement is reported separately as

$$D_{\theta,s} = \frac{\|\theta_s - \theta_s^{\mathrm{KD}}\|_2}{\|\theta_s^{\mathrm{KD}}\|_2}. \tag{32}$$

These post-training quantities diagnose the realized intervention; they are not comparator-identification conditions.

An MDE describes the resolution of a planned contrast under its declared variance model, sample size, error rate, and power. It does not show that a smaller effect is zero or that a larger effect matters biologically. Likewise, language-quality criteria decide whether two arms are sufficiently comparable for a specified analysis; they do not establish a biological endpoint.

The $+0.002$ unique-R^2 rule was fixed before the transfer results were seen, to decide whether participant-level biological transfer justified a matched-control follow-up, and was then kept unchanged as the reference for the target-projection assay. It can support a statement about whether an assay-specific simultaneous bound reaches that continuation rule. It cannot define biological importance across another response scale, nuisance model, layer, cohort, or target.

D.4 Information-theoretic boundaries

Population partial R^2 has an information-theoretic identity only under restrictive assumptions. For scalar Y, jointly Gaussian (X, Y, Z), correctly specified population-linear conditional means, and unregularized population residual variances,

$$I(X; Y \mid Z) = -\frac{1}{2}\log\bigl(1 - R^2_{\mathrm{partial}}\bigr), \qquad \Delta\mathcal{R}^2_{\mathrm{semi}} = \bigl(1 - \mathcal{R}^2_N\bigr)\, R^2_{\mathrm{partial}}. \tag{33}$$

Here $\Delta\mathcal{R}^2_{\mathrm{semi}} = \mathcal{R}^2_F - \mathcal{R}^2_N$ is the population analogue of the cross-validated increment defined above. Equation 33 converts a population residual-variance ratio into conditional mutual information under the stated Gaussian model. The multivariate analogue uses a log-determinant ratio. The thesis instead estimates finite-sample, cross-validated semipartial differences from separately regularized ridge fits. Those differences may be negative and depend on the nuisance design, rank, penalty, and held-out substrate, so they are neither estimates nor lower bounds of $I(X; Y \mid Z)$ [32].

A data-processing claim requires a separate conditional-independence premise. Let S be stimulus text, B a recorded brain response, $T = g(S)$ the deterministic synthetic brain-response target, and H the retained student state. The chain

$$B \longrightarrow T \longrightarrow H \qquad \Longleftrightarrow \qquad B \perp H \mid T$$

would require all information from B that reaches H to pass through T, which would imply $I(B; H) \leq I(B; T)$. The executed design does not establish this chain because S directly supplies both T and H, and it also drives B. Conditioning on S changes the question. Because T is deterministic given S, $I(T; B \mid S) = 0$. That identity does not show that the target is useless. It shows that conditioning on the exact stimulus cannot establish additional target information beyond that stimulus or recover the desired unconditioned mediation claim. Data processing therefore supplies a bound only after the relevant conditional independence is defended; it does not predict the sign of a held-out encoding contrast [32].

Output matching and auxiliary-target fitting impose still weaker constraints on retained representations. Let p^τ_{tch} and q^τ_{stu} be the teacher and student output distributions at softmax temperature τ, let H_{tch} and H_{stu} be their internal states, let d_H be any declared discrepancy between those states, and let $\Delta R^2_{\mathrm{transfer}}$ be the held-out biological-transfer contrast on recorded responses. Then, without additional identifying assumptions,

$$D_{\mathrm{KL}}(p^\tau_{\mathrm{tch}} \| q^\tau_{\mathrm{stu}}) = 0 \;\not\Rightarrow\; d_H(H_{\mathrm{stu}}, H_{\mathrm{tch}}) = 0, \qquad \mathcal{L}_{\mathrm{target}} \downarrow \;\not\Rightarrow\; \Delta R^2_{\mathrm{transfer}} > 0.$$

The first non-implication holds because different internal parameterizations can realize the same output distribution. The second holds because a temporary target map can absorb part of the auxiliary objective, and a predictable target can be learned without producing participant-general improvement on recorded responses. Low output or target loss can establish optimization success under its own objective. It cannot replace fresh tests of retained-student movement, comparator adequacy, brain-response attribution, biological transfer, or practical utility. Section 4 reports those empirical stages without treating any formal analogy as their outcome.

E Supporting Analyses and Stopped Routes

This appendix separates secondary evidence from work that did not reach an interpretable scientific test. A working pipeline, a restricted local diagnostic, and a stopped route answer different questions. Appendix A states how far each record's conclusion may be taken, and Appendix F reports the full numerical results and sensitivity analyses. The headline empirical verdicts remain in Section 4.

E.1 Pipeline validation and scientific scope

The end-to-end pipeline was first tested in a setting where the expected signal was known. Real Pereira sentences [28] were paired with synthetic responses generated from teacher features, named nuisance variables, and noise. The test exercised ordinary and target-guided KD, the temporary target head, middle-layer extraction, the contiguous held-out split, fold-local ridge fitting, nuisance subtraction, and aggregation across optimization seeds. The expected teacher–student ordering appeared, the nuisance component remained small, and excessive target weight degraded the result rather than receiving an automatic advantage.

These observations rule out several basic implementation failures. They show that the held-out evaluation tail was not used for target-guided training, that the scoring pipeline could recover a planted feature–response relation, and that the analysis did not reward every target-guided run by construction. They do not establish that an LM predicts recorded brain responses, that the nuisance set is sufficient for real stimuli, or that a brain-response objective improves a student. The synthetic test therefore validates the end-to-end implementation, not the scientific claim.

E.2 Response averaging and recorded-response robustness

Two averaging analyses addressed different questions. The first changed how many participants contributed to the recorded-response intervention target. Because each target was rebuilt from a different participant subset, increasing the subset size also changed participant identity, response reliability, and target construction. A corrected analysis showed that the apparent gain in controlled brain predictivity did not rise steadily with subset size. The result therefore does not identify a causal benefit of adding more participants to the training target.

The second analysis held the frozen representation and encoding procedure fixed while averaging increasing numbers of recorded responses. Controlled brain predictivity increased as averaging suppressed idiosyncratic and measurement noise and emphasized the shared stimulus-evoked component. This is a valid signal-to-noise gain for a group-response estimand. It changes what is being predicted, however, and does not supply participant replication or rescue the individual-response intervention.

The remaining analyses tested whether specific design choices concealed a recorded-response benefit. Table 14 maps each change to its objection and scope.

These analyses weaken the listed specific alternative explanations. They do not establish that every recorded-response objective must fail. Section 4.3 states the participant-level intervention conclusion, and Appendix F reports the numerical variants.

E.3 Reduced external and cross-modal probes

External protocols were used as local diagnostics only when their decisive ingredients could be approximated with the available data and models. The standard is therefore narrower than reproduction. The question is whether a reported effect appears under the local conditions.

The first diagnostic adapted two elements of Negi et al., a differentiable head that aligns model features to response timing and their controls for target specificity [9], to the available LeBel responses [25] and decoder-only LMs. Under the gentlest tested training schedule the student representations that remain after the temporary head is discarded changed little, and stronger schedules degraded language quality severely. The local runs therefore did not establish a target-specific manipulation over their controls. The comparison nevertheless differs from the source study in population, response substrate, model family, language, training scale, and control construction. It is therefore a restricted local failure to establish the target-specific manipulation, not a reproduction or refutation of the published result.

Table 14: Recorded-response robustness analyses and their scope.

Objection	Design change	Supported disposition	Remaining open
Auxiliary weight was poorly chosen.	Repeated the recorded-response intervention over the tested weight range.	Only the highest tested weight gave a fold-level interval against the matched permuted control that excluded zero, and at a substantial language-quality cost, so no quality-matched target-specific gain is supported.	Other objectives or schedules could behave differently.
Adapter capacity was too limited.	Increased the tested low-rank adaptation capacity.	The tested capacity increase did not recover the effect.	This does not test every stronger manipulation.
Pointwise regression was the wrong loss.	Replaced it with a contrastive response objective.	The alternative loss did not produce a reliable target-specific benefit.	Other response objectives remain possible.
Regional targets hid a voxelwise effect.	Tested the available naturalistic voxelwise training procedure.	Target-matched tuning was neutral at the lowest tested target weight and degraded held-out predictivity at higher weights, never exceeding either the untuned model or its permuted twin, so the intended change was never induced.	The broader multi-participant full-parameter route was not built.
Parameter-efficient tuning was too restrictive.	Tested full-parameter tuning, including a gentler three-participant probe.	Aggressive tuning caused forgetting; the gentler probe showed no reliable target-specific gain.	Three participants do not support population inference.

The second diagnostic adapted a speech-based protocol from Moussa et al. [8]. Starting from the same Wav2Vec2-base model family, the design required a positive phoneme-prediction gain before constructing a matched-quality specificity test. In the reduced single-participant setup, target loss decreased but the phoneme-gain criterion was not met, so the downstream specificity comparison was not built. The source study used a larger, multi-participant setting with different timing and model choices. The result is thus a reduced local non-reproduction of the prerequisite effect, not evidence against the published claim.

Neither diagnostic supplied an external positive control for the present intervention pipeline. They narrow what the local pipeline recovered while leaving the external literature open under its original conditions.

E.4 Failed instruments and precondition-stopped routes

Three routes stopped because an upstream measurement or attribution condition failed. In each case, continuing to intervention would have produced a number without a defensible interpretation.

The shared-response route asked whether the predictable group response could serve as a ceiling for participant-specific intervention effects. In principle, this reference estimates the expected response shared across participants for the same stimulus, $m(S) = \mathbb{E}[Y \mid S]$. In practice, the leave-one-participant reference was not reliable enough, while temporally adjacent samples preserved apparent residual predictivity through leakage and autocorrelation. Once a clean temporal gap was imposed, the controlled residual predictivity was small but too imprecise to fix a ceiling for participant-specific effects. The ceiling instrument therefore failed; the analysis does not show that participant-specific structure is absent or that a universal biological ceiling has been reached.

The reading-time route asked whether a cognitive target could provide a simpler intervention signal than fMRI. Training a model with an auxiliary head on residualized reading times initially improved how efficiently a probe on the frozen representation learned held-out text, by more than any control whose target was matched for learnability. A phase-randomized control preserved the target's marginal distribution and temporal structure, however, and matched that improvement. The design therefore did not isolate cognition-specific content from target shape. This attribution failure does not imply that reading time contains no useful signal; it means that the available contrast could not identify that signal.

The gaze route required three conditions before privileged-information training. It needed a measurable target, evidence that it added information beyond text, and enough independent groups for leakage-resistant evaluation. Natural-reading gaze was measurable but did not add information beyond the text itself under the tested assay. Task-directed gaze was more predictable, but the signal could reflect task intent or label leakage, and the number of independent paragraph groups was too small for a reliable group split. The intervention therefore stopped at its preconditions. The route's other planned signal, electroencephalography (EEG), never reached testing and supplies no evidence.

Stopping at these failed prerequisites means no downstream number exists that could be misread as an identified intervention effect. They identify why the proposed instruments could not answer their questions, but they do not turn measurement failure into a biological null.

F Claim-Relevant Sensitivities and Verification Tables

This appendix reports the verification detail needed to judge whether the main conclusions depend on an analysis choice or an unstable aggregate. Section 4 states the decisive observations and scientific verdicts, and Appendix A maps each of them to the experiment that settled it.

The tables preserve the inference-unit distinction defined in Appendix D. Participant-level effects support inference across the sampled biological units, whereas held-out blocks diagnose dependence on the fixed stimulus partition and optimization seeds diagnose the stability of a fixed training procedure.

F.1 Claim-relevant analysis choices

An analysis choice can affect a thesis claim in four ways. An *interpretation-changing* analysis replaces an earlier estimand or inference unit. A *scope-bounding* sensitivity narrows what the primary result supports without reversing its frozen decision rule. A *confirmatory* check leaves the scoped interpretation unchanged. A *limitation or correction* removes authority from a design element rather than generating a more favorable result. Tables 15 and 16 apply these labels to the analyses that matter for the thesis claims.

The interpretation-changing analysis altered the question being answered rather than reanalysing the earlier averaged-target result. The remaining sensitivities either narrowed the scope, confirmed stability near a frozen specification, or exposed a limitation.

F.2 Participant-level intervention contrasts

The participant-specific recorded-response estimator first forms a paired target-specific contrast for participant u, held-out block k, and optimization seed s, where $P = 5$ is the number of

Table 15: Interpretation-changing and scope-bounding analyses.

Analysis choice	Scientific role	Status	Effect on interpretation
Representation rank	Distillation-headroom analysis	Scope-bounding	The ordering of the model arms survived, but absolute magnitudes and some signs changed. Headroom is conditional on the declared rank.
Auxiliary weight and language quality	Participant-averaged recorded-response intervention	Scope-bounding	Near-quality-matched settings remained uncertain across folds. The clearest fold-positive setting incurred a substantial language-quality cost.
Quality band and model family	Distillation-headroom analysis	Scope-bounding	The full-range Pearson correlation between log bits per byte and unique R^2 was -0.783, with interval [-0.911, -0.500]; within the capable-model band it was -0.501, with interval [-0.859, $+0.188$]. Language quality remains a required control, not a deterministic explanation.
Response construction and inference unit	Participant-specific recorded-response intervention	Interpretation-changing	Participant-specific targets and participant-first inference answer a different question from an averaged target and fold-level summary. The change removed support for a participant-general gain.
Participant and subgroup deletion	Saved-student biological-transfer assay and 50-component linear target-projection assay	Scope-bounding	Removing the predeclared highest-baseline participant reversed the relative transfer point estimate. Every leave-one-participant-out linear target-projection mean remained negative.
Held-out stimulus block	Controlled naturalistic voxelwise predictivity assay; recorded-response intervention; saved-student biological-transfer assay; 50-component linear target-projection assay	Scope-bounding	The naturalistic assay remains a one-participant result. No leave-one-block-out intervention analysis produced a positive participant-mean contrast that changed the participant-level conclusion, and the linear target-projection result held under every block deletion.
Controlled-predictivity contrast by seed	Practical-utility evaluation	Scope-bounding	Only $5/8$ of the eight technical-seed signs were positive. The utility analysis therefore follows a seed-unstable prerequisite rather than a reliably induced target-specific change.

matched permuted-target draws.

$$d_{uks} = R^{2,\text{unique}}_{\text{recorded},uks} - \frac{1}{P}\sum_{p=1}^{P} R^{2,\text{unique}}_{\text{perm}(p),uks}, \qquad e_u = \text{median}_{k,s}\, d_{uks}, \qquad \widehat{\theta}_{\text{part}} = \frac{1}{U}\sum_{u=1}^{U} e_u. \tag{34}$$

Equation 34 subtracts the mean matched permuted-target score inside every seed–block cell, then aggregates the crossed cells within participant before averaging participants. The resulting U =9 participant effects, each based on 15 paired cells, are the biological inputs to the cohort estimate.

The signs are mixed around a small cohort mean. The table supports participant-first aggregation and heterogeneity inspection; it does not imply that every participant has exactly zero effect.

The saved-student biological-transfer assay uses a different target and within-participant aggregator. Let T_u, X_u, and K_u denote the participant-level scores for the synthetic-response target, text-derived

Table 16: Confirmatory checks and design limitations.

Analysis choice	Scientific role	Status	Effect on interpretation
Layer, nuisance set, and target aggregation	Saved-student biological-transfer assay and 50-component linear target-projection assay	Confirmatory	Prespecified nearby choices did not overturn the primary rules. The 50-component linear target projection did not meet the continuation rule fixed for it in advance (Section 4.4); retention and composed-path quantities remain diagnostics of that pathway.
Saved-student optimization seed	Synthetic-target recovery assay; retained-student movement audit; saved-student target-retention analysis; saved-student biological-transfer assay; composed-path diagnostic	Confirmatory	The seeds agreed where the manipulation itself was measured, with target extraction positive and the movement criterion met in all six, and disagreed in sign on the endpoint contrasts for incremental retention, biological transfer, and the composed path. Seed agreement can establish reproducible optimization or manipulation, but it cannot substitute for participant agreement.
Saved controls audited after training	Comparator-adequacy audit and attribution assessment	Limitation or correction	The saved block-permutation sensitivity is defective, and the text-derived auxiliary control fails the comparability checks made before outcomes were inspected, including comparability with the ordinary-KD baseline. Neither can identify an effect of brain-response content.

auxiliary control, and ordinary KD. The three displayed contrasts satisfy

$$\Delta_u^{T-X} = T_u - X_u, \quad \Delta_u^{T-K} = T_u - K_u, \quad \Delta_u^{X-K} = X_u - K_u, \quad \Delta_u^{T-X} = \Delta_u^{T-K} - \Delta_u^{X-K}. \tag{35}$$

Equation 35 shows why a positive relative contrast against the auxiliary control need not imply improvement over ordinary KD, because the auxiliary control may instead have moved downward. Seeds and outer folds are averaged within each participant before these contrasts are formed.

The transfer contrasts are also heterogeneous. The relative mean Δ^{T-X} is small and positive, while the direct mean Δ^{T-K} is -0.000014, with interval $[-0.000739, +0.000711]$; removing the participant with the highest ordinary-KD brain-predictivity baseline, predeclared for deletion, changes the relative point estimate to -0.000131. These rows explain why the relative comparator cannot replace the direct contrast or participant-first inference.

The two participant tables must not be pooled or compared row by row. They use different targets, contrasts, and within-participant aggregators; the shared identifiers only show that they were evaluated in the same participant cohort.

Table 17: Comparator-audit checks classified as identification checks or post-training diagnostics.

Role	Total	Pass or within tolerance	Fail or outside tolerance	Interpretation
Identification checks	37	20	16	One additional check is unresolved; failures include KD-baseline recovery and headroom, covariance geometry, effective rank, and nuisance predictability.
Post-training diagnostics	18	4	14	Parameter, Frobenius, and CKA margins describe realized intervention differences and do not determine comparator validity.

Table 18: Participant-specific recorded-response intervention contrasts.

UID	Participant effect e_u (unique R^2 difference)
797	$+0.00002885$
837	$+0.00100412$
841	-0.00018259
848	-0.00063374
856	-0.00037809
865	-0.00058994
875	$+0.00070575$
876	$+0.00011067$
880	$+0.00083826$
Mean	$+0.00010$

Table 19: Participant-level saved-student biological-transfer contrasts.

UID	Δ_u^{T-X}	Δ_u^{T-K}	Δ_u^{X-K}
797	-0.00023801	-0.00005311	$+0.00018490$
837	$+0.00227168$	$+0.00226591$	-0.00000577
841	$+0.00000452$	-0.00078813	-0.00079265
848	-0.00002346	-0.00002667	-0.00000320
856	-0.00022167	-0.00036255	-0.00014087
865	-0.00064994	-0.00108762	-0.00043767
875	$+0.00009116$	-0.00006339	-0.00015455
876	-0.00013023	-0.00014342	-0.00001319
880	$+0.00012305$	$+0.00013517$	$+0.00001212$
Mean	$+0.000136$	-0.000014	-0.000150

F.3 Participant-level mechanism diagnostics

The target-projection diagnostic evaluates the declared 50-component linear pathway on the same sentence substrate; it is not a prerequisite for the independent direct-transfer contrast. For each participant, A_u measures the target's unique response predictivity above nuisance, and Q_u compares that score with the frozen row-twin control, which keeps the target rows and the readout procedure but breaks their pairing with the sentences (Section 4.4). These quantities compare target-above-

Table 20: Participant-level linear target-projection diagnostics.

UID	Target above nuisance A_u	Aligned minus frozen twin Q_u
797	-0.00266759	-0.00031777
837	-0.00448823	-0.00785204
841	-0.02042083	$+0.00588154$
848	-0.00001541	$+0.00158126$
856	-0.01394534	-0.00066703
865	$+0.00208088$	$+0.00684375$
875	$+0.00266981$	$+0.01519279$
876	-0.00660269	-0.00590127
880	$+0.00293994$	$+0.00106256$
Mean	-0.004494	$+0.001758$

Table 21: Participant-level composed-path quantities.

UID	Absolute overlap O_u	Synthetic response minus KD D_u	Aligned minus twin J_u
797	$+0.00045651$	$+0.00018799$	$+0.00019106$
837	$+0.00484230$	$+0.00029143$	$+0.00026774$
841	$+0.00038073$	$+0.00004280$	$+0.00003954$
848	$+0.00010189$	-0.00005708	-0.00005644
856	-0.00040275	-0.00005387	-0.00005261
865	$+0.00216009$	-0.00006632	-0.00005151
875	$+0.00447180$	-0.00018830	-0.00019123
876	-0.00126093	$+0.00008691$	$+0.00008438$
880	$+0.00043030$	$+0.00003540$	$+0.00003336$
Mean	$+0.001242$	$+0.000031$	$+0.000029$

nuisance predictivity with sentence-pairing specificity. They are read before the later target-retention analysis and composed-path diagnostic.

The mean target-above-nuisance score is -0.004494, with two-sided 95% t interval $[-0.010695, +0.001707]$ and 3 of 9 participant increments positive under an exact sign test at 0.5078, and every leave-one-participant-out mean remains negative. That score does not meet the continuation rule fixed for it in advance (Section 4.4). The aligned-minus-frozen-twin values have substantial effects of both signs, so the paired specificity question remains unresolved. It does not contradict the first result.

The later diagnostics ask whether the part of the target that a student predicts overlaps with the target direction that predicts participant responses. For participant u, O_u is the absolute composed student-to-target-to-response overlap, D_u is its brain-target-minus-KD increment, and J_u is the aligned-minus-frozen-twin increment. All three average optimization seeds and outer folds within participant before participant-level inspection.

Absolute composed overlap remains unresolved. The two increments are smaller and have mixed signs. Within the declared linear pathway, these downstream diagnostics cannot repair the target-projection result or identify its cause. They do not constrain the independent direct-transfer contrast.

F.4 Technical-seed diagnostics

Optimization seeds test whether a fixed training procedure behaves reproducibly. They do not support biological generalization, because the same participants and stimuli are reused within each seed.

The practical-utility evaluation has a prerequisite, a reliable contrast in controlled brain predictivity between the recorded-response arm and its matched permuted-response control. Its brain-specific interpretation depends on that contrast. Table 22 reports the complete seed vector used to judge that prerequisite contrast; it declared eight seeds, where the transfer and target-extraction procedures declared six.

The signs are split, and the mean is strongly influenced by seed 0. The seed rows are rounded independently to three decimal places, whereas the mean and median are computed from the un-

Table 22: Technical-seed values for the practical-utility prerequisite.

Seed	Recorded-minus-permuted contrast in controlled brain predictivity
0	$+0.063$
1	$+0.014$
2	$+0.004$
3	$+0.016$
4	$+0.005$
5	-0.006
6	-0.002
7	-0.005
Mean	$+0.011$
Median	$+0.0042$

Table 23: Technical-seed values for saved-student biological transfer.

Seed	Δ^{T-X}	Δ^{T-K}	Δ^{X-K}
0	-0.00015727	-0.00035769	-0.00020042
1	$+0.00032614$	$+0.00003556$	-0.00029058
2	$+0.00004025$	$+0.00000892$	-0.00003133
3	$+0.00025080$	$+0.00015307$	-0.00009773
4	$+0.00018344$	-0.00006376	-0.00024721
5	$+0.00017470$	$+0.00014137$	-0.00003333
Mean	$+0.000136$	-0.000014	-0.000150

Table 24: Technical-seed target-extraction and retention values.

Seed	Target extraction M_s	Incremental retention B_s
0	$+0.03965524$	$+0.00013878$
1	$+0.04096723$	$+0.00103098$
2	$+0.03932328$	-0.00159873
3	$+0.04291468$	-0.00234707
4	$+0.03920675$	-0.00038085
5	$+0.04228561$	-0.00197014
Mean	$+0.040725$	-0.000855

rounded values. The complete vector documents mixed signs and seed-0 sensitivity; it does not establish a null or equivalence.

For the saved-student biological-transfer assay, each seed row averages participants and outer folds. The three columns use the same contrasts as Equation 35.

Before participant responses were inspected, the synthetic-target recovery, retained-student movement, and language-quality criteria were satisfied. The largest saved-arm language-quality difference, 0.002579, remained within the frozen 0.005 margin. The seed table therefore checks an interpretable manipulation while leaving biological inference to the participant table.

The exact-target seed diagnostics separate absolute target extraction from improvement over seed-matched ordinary KD and from composed-path increments. Within each seed, target extraction and incremental retention aggregate held-out target recovery; the two overlap increments average participants and outer folds.

Absolute target extraction is positive in every seed. Incremental retention has mixed signs and an interval containing zero. Both composed-path increments show the same uncertainty. The separation shows that fitting the target in absolute terms does not establish improvement over ordinary KD or transfer along the composed path.

The values use layer 6 and the same 1,999 WikiText sentences held out from every training arm; Equation 31 defines both distances. Before participant responses were opened, every seed was

Table 25: Technical-seed composed-path increments.

Seed	Synthetic response minus KD D_s	Aligned minus twin J_s
0	-0.00003781	-0.00004017
1	-0.00004152	-0.00004668
2	-0.00001666	-0.00001488
3	$+0.00006481$	$+0.00006414$
4	$+0.00014778$	$+0.00014581$
5	$+0.00006938$	$+0.00006797$
Mean	$+0.000031$	$+0.000029$

Table 26: Movement of the retained synthetic-target student away from its seed-matched ordinary-KD student, by technical seed.

Seed	Centered relative-Frobenius	Linear-CKA distance
0	0.1864	0.0026
1	0.2163	0.0055
2	0.2987	0.0118
3	0.1955	0.0036
4	0.1971	0.0039
5	0.1965	0.0034

Table 27: Models in the cross-family language-quality analysis.

Model	Family	Parameters	Mid layer	Bits/byte	Unique R^2
Mistral-7B-v0.1	Mistral	7.24B	16	1.135	+0.0438
Qwen2.5-7B	Qwen	7.62B	14	1.152	+0.0235
Llama-3.2-3B	Llama	3.21B	14	1.163	+0.0464
Qwen2.5-3B	Qwen	3.09B	18	1.200	+0.0371
OPT-6.7B	OPT	6.66B	16	1.222	+0.0233
Pythia-2.8B	Pythia	2.78B	16	1.240	+0.0174
Qwen2.5-1.5B	Qwen	1.54B	14	1.242	+0.0297
Llama-3.2-1B	Llama	1.24B	8	1.249	+0.0360
OPT-2.7B	OPT	2.65B	16	1.276	+0.0324
Pythia-1.4B	Pythia	1.41B	12	1.283	+0.0213
Pythia-1B	Pythia	1.01B	8	1.315	+0.0226
OPT-1.3B	OPT	1.32B	12	1.315	+0.0281
Qwen2.5-0.5B	Qwen	0.49B	12	1.342	+0.0356
GPT-2 Large	GPT-2	0.77B	18	1.351	+0.0173
Pythia-410M	Pythia	0.41B	12	1.381	+0.0233
GPT-2 Medium	GPT-2	0.35B	12	1.394	+0.0177
OPT-350M	OPT	0.33B	12	1.453	+0.0259
GPT-2 Small	GPT-2	0.12B	6	1.486	+0.0170
OPT-125M	OPT	0.13B	6	1.514	+0.0147
Pythia-160M	Pythia	0.16B	6	1.523	+0.0209
DistilGPT-2	GPT-2	0.08B	3	1.628	+0.0071
Pythia-70M	Pythia	0.07B	3	1.699	-0.0012

required to exceed 10^{-4} in centered relative-Frobenius change and to exceed the larger of 10^{-6} and ten times the duplicate-extraction noise in linear-CKA distance. Because repeated extraction of the same checkpoint reproduced it exactly, 10^{-6} was the binding floor. All six seeds cleared both criteria.

Unique R^2 is the variance explained above the declared nuisance set, and all 22 models enter the primary middle-layer correlation. The capable-model sensitivity retains the ten models below 1.30 bits per byte. The family-factor analysis excludes Llama and Mistral because each has fewer than three model sizes; those exclusions do not apply to the primary cross-family correlation.

These tables distinguish choices that changed or bounded an interpretation from checks of local stability.

Data and code availability

The Tuckute condition-B responses [24], the LeBel story-listening responses [25], the Pereira sentence set [28], WikiText-103 [26], and TRIBE v2 [27] are publicly available from their original sources. The analysis code, the numbered experiment records E001 to E030, and the source of this manuscript are available at `https://github.com/MohammadErfan-Jabbari/brain-alignment`. Appendix C states the reproduction scope and the limits on exact replay.

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